DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
INDIAN INSTITUTE OF TECHNOLOGY MADRAS
CHENNAI – 600036

Cryptographic applications of k-SUM

A Thesis

Submitted by

AKHIL VANUKURI

For the award of the degree

Of

MASTER OF SCIENCE

July 2024

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THESIS CERTIFICATE

This is to undertake that the Thesis titled **CRYPTOGRAPHIC APPLICATIONS OF K-SUM**, submitted by me to the Indian Institute of Technology Madras, for the award of **Master of Science**, is a bona fide record of the research work done by me under the supervision of **Dr. Shweta Agrawal**. The contents of this Thesis, in full or in parts, have not been submitted to any other Institute or University for the award of any degree or diploma.

Chennai 600036 **Akhil Vanukuri**

Date: May 2024

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LIST OF PUBLICATIONS

I. PUBLICATIONS IN CONFERENCE PROCEEDINGS

1. Shweta Agrawal, Sagnik Saha, Nikolaj Ignatieff Schwartzbach, Akhil Vanukuri, Prashant Nalini Vasudevan. k-SUM in the Sparse Regime: Complexity and Applications. In CRYPTO, 2024. (To appear).

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ABSTRACT

KEYWORDS k-sum, average case hardness, public key encryption, somewhat homomorphic encryption, extractable commitment scheme, collision resistant hash function

In the average-case k-SUM problem, given r integers chosen uniformly at random from $\{0, \ldots, M-1\}$, the objective is to find a "solution" set of k numbers that sum to 0 modulo M. In the *sparse* regime of $M \gg r^k$, our work [ASS+23] initiates the study of k-SUM and its variant k-XOR, especially their *planted* versions, where a random solution is planted in a randomly generated instance and has to be recovered. In that paper we provide evidence for the hardness of these problems and show applications to cryptography by constructing a public-key encryption scheme (PKE) based on LPN and k-XOR. In this thesis, we discuss some of the work mentioned in [ASS+23] and extend it. The summary of the results in this thesis is as follows:

- **Public Key Encryption**. We give the first Public Key Encryption (PKE) scheme construction assuming $2^{\frac{m}{\log \frac{mq}{B}}}$ hardness of the Learning With Error (LWE) problem[1] with any super constant modulus to noise ratio (i.e., $\frac{q}{B} = \omega(1)$) along with assuming mild k-SUM hardness. Additionally, we show that assuming mild k-SUM hardness we can construct Public Key Encryption (PKE) from a weaker variant of the LWE problem than was known before. In particular, such LWE hardness does not appear to imply PKE on its own – this suggests that k-SUM can be used to bridge "minicrypt" and "cryptomania" in some cases, and may be applicable in other settings in cryptography.

- **New reductions and algorithms**. We define new variants of the k-SUM problem to use in the construction of the above said PKE scheme and reduce those variants to the original k-SUM problem. This allows us to use mild hardness of k-SUM

[1] Assuming LWE is $T(r)$ hard informally means assuming that any algorithm that runs in time less than $T(r)$ cannot solve LWE with given parameters

problem to build the above PKE scheme. Additionally, we present a new algorithm for average-case k-XOR that is faster than known worst-case algorithms at low densities.

- **Somewhat Homomorphic Encryption**. We introduce a new problem called planted probabilistic error LPN problem ($\mathsf{PPE-LPN}$) which can be viewed as a simple modification to the LPN problem. Assuming the hardness of the $\mathsf{PPE-LPN}$ problem, we build a Somewhat Homomorphic Encryption (SHE) scheme. Our SHE is interesting because many attempts to build Fully Homomorphic Encryption (FHE) from assumptions other than LWE have failed. While even our scheme does not achieve fully homomorphic encryption, it can be looked at as a technical contribution towards a possible realisation of an LPN based (or any non-lattice problem based) FHE.

- **Others**. We notice that we can construct extractable commitment schemes using the “minicrypt" LPN and k-XOR based PKE scheme of [ASS+23]. This implies that we get a Universally Composable Oblivious Transfer (UC-OT) protocol due to the results stated in [DDN14]. Intriguingly, the level of hardness needed from LPN in the above-said UC-OT protocol was previously not known to give a UC-OT protocol by itself. We also examine cryptographic applications of k-SUM (k-XOR) in the dense regime by constructing a Collision Resistant Hash Function (CRHF) assuming the hardness of a variant of k-SUM in the dense regime. This construction contributes to the theme of building cryptographic primitives from diverse hardness assumptions. This is useful because technological or algorithmic advancements can break the existing hardness assumptions.

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CHAPTER 1

OVERVIEW

In the k-SUM problem, given a set of r numbers from $\{0, \ldots, M-1\}$, the task is to find a set of k of them that sum to 0 (modulo M), if such a set exists.[1] This problem has been central to studying the complexity of important problems in a variety of domains such as computational geometry, data structures and graph theory [AW14, Pat10, GO95, BHP01, SEO03, KPP16]. It has found several applications in cryptanalysis [Wag02, BCJ11], and is essentially a parameterized version of the Subset Sum problem. In this work, based on the theory developed in [ASS+23], we explore the cryptographic applications of the k-SUM problem.

One reason for looking at problems like k-SUM or its variants as a source of hardness in cryptography is to find reasonable computational assumptions that are new and incomparable to existing assumptions. There is currently very little variety in the problems whose hardness is being used to construct advanced cryptographic primitives. If our assumptions rely on a different kind of hardness than existing ones, they would represent a fallback in case some of the other family of assumptions break down due to algorithmic or technological developments.

Average-Case k-SUM. In the *average-case* k-SUM problem, the r numbers in the input are each chosen uniformly at random from $\{0, \ldots, M-1\}$. The characteristics of the problem now change depending on the relative sizes of M, r, and k. In analogy to

[1] The k-SUM problem is usually defined with the sum being over the integers. These variants are equivalent in complexity in the worst-case and also in the average-case in certain regimes of parameters (see [BSV21, DKK21]).

Subset Sum [LO85], we define the *density* of average-case k-SUM as the following ratio:

$$\Delta = \frac{\log \binom{r}{k}}{\log M}.$$

When this density Δ is 1, the expected number of k-SUM solutions in an instance is also 1. In general, the expected number of solutions is approximately $r^{k\,(1-1/\Delta)}$. For values of Δ larger than 1 (the *dense* regime), this number is polynomial in r, and for Δ smaller than 1 (the *sparse* regime), the number of solutions is vanishing with r. When k is small, the density can be approximated by the ratio $(k \log r / \log M)$, and we will use this simplification in the remainder of this work[2].

In the dense regime, several non-trivial algorithms are known for average-case k-SUM that are more efficient than the worst-case algorithms. For instance, the "birthday" algorithm that computes the sum of random sets of $k/2$ numbers and looks for collisions runs in expected time $O(\sqrt{M}) = O(r^{k/(2\Delta)})$ which is $r^{O(1)}$ for densities larger than k. For densities larger than $\approx k/(1 + \log_2 k)$, Wagner's k-tree algorithm [Wag02] solves the problem in time $\widetilde{O}(r)$.[3] However, at density $\Delta = 1$, when there is only a single solution in expectation, no algorithms are known that outperform the best worst-case algorithms. The average-case k-SUM problem at this density is believed to be as hard as the worst-case variant, although no worst-case to average-case reduction is currently known [Pet15, LLW19, DKK21].

Conjecture 1.1 (Average-Case k-SUM Conjecture [Pet15, LLW19, DKK21])**.** *Any algorithm for average-case k-SUM at density* 1 *with constant probability of success has running time at least* $\Omega\left(r^{\lceil k/2\rceil - o(1)}\right)$.

In the sparse regime, little was known before [ASS+23] initiated the investigation of the

[2]Please see [ASS+23] for a discussion on the accuracy of this approximation.
[3]This algorithm was used by Wagner to perform non-trivial attacks on a variety of cryptosystems.

average-case complexity of k-SUM at densities less than 1, where random instances are unlikely to have solutions. In this regime where solutions are unlikely, a meaningful variant of the k-SUM problem that they consider is the *planted* k-SUM problem. Here, a randomly chosen set of k numbers that sum to 0 is planted at random locations in a random k-SUM instance. Two problems arise naturally in this setting:

- **The *planted search* k-SUM problem** is to recover a k-SUM solution given such an instance (at low densities, with high probability, the planted solution is the only one).

- **The *planted decision* k-SUM problem** distinguishes a random instance with a planted solution from a random instance without a planted solution.

As part of their exploration of the k-SUM problem, [ASS+23] give a construction of Public-Key Encryption (PKE) from somewhat mild hardness of planted search k-XOR (a variant of the k-SUM problem) at sufficiently low densities, assuming weaker hardness of the Learning Parity with Noise (LPN) problem than was known before. Before this, we only knew how to construct PKE assuming either that LPN with m-bit secrets at noise rate $O(1/\sqrt{m})$ is hard for $\mathsf{poly}(m)$-time algorithms [Ale03b], or that LPN with constant noise rate is hard for $2^{m^{0.5}}$-time algorithms [YZ16]. In contrast, [ASS+23] shows that adding k-XOR enables us to use just 2^{m^c} hardness (for any constant $c > 0$) of LPN with constant noise rate. More formally, their theorem statement is as follows:

Theorem 1.2 (PKE from k-XOR and LPN [ASS+23])**.** *There are values of $\eta = \Theta(1)$, $k = O(\log r)$, $m = r^{o(1)}$, and $\ell = \mathsf{poly}(r)$ such that the cryptosystem $PKE_{\eta,k,\ell,m}$ is a (weak) public-key bit encryption scheme, with all operations running in time $r^{1+o(1)}$, if any of the following conditions are satisfied:*

1. *Constant-noise LPN is 2^{m^c}-hard (for any constant $c > 0$) and planted search k-XOR is $r^{k/2-o(1)}$-hard[4] at every density $\frac{1}{\mathsf{polylog}(r)}$;*

[4]Informally, it means that no algorithm running in time $r^{k/2-o(1)}$ can solve the k-XOR problem at the

An interesting aspect about the above PKE-scheme is that the LPN hardness (or) the k-XOR hardness assumed by it don't seem to imply PKE by themselves. This suggests the possibility of the k-SUM family of problems serving as a bridge for problems from the world of "Minicrypt" (where one-way functions exist) to the world of "Cryptomania" (where public-key encryption exists) — see also [Imp95].

1.1 OUR RESULTS

Public Key Encryption. We begin this work by extending the ideas of [ASS+23] and constructing a PKE scheme (Section 4.2) using a weaker LWE assumption (which by itself is not known to imply PKE) along with the hardness of k-SUM (which, again by itself does not imply PKE). We show the following:

Theorem 1.3 (PKE from LWE). *For any security parameter $r \in \mathbb{N}$ and error rate $\alpha \leq \frac{1}{\log \log r}$ there exists values $m = \log r \cdot \omega(1)$, $k = \omega(1)$, $q \geq 2\sqrt{m}\,k$ such that assuming the corresponding Learning With Error (LWE) problem with parameters r, m, k, α to be $2^{\frac{m}{\log \frac{m}{\alpha}}}$[5]-hard gives us a PKE scheme that satisfies weak* IND-CPA *security (Definition 4.1).*

To the best of our knowledge, before this, we only knew how to construct PKE assuming that LWE with m-bit secrets at an error rate $\tilde{O}(\frac{1}{\sqrt{m}})$ is hard for $\mathsf{poly}(m)$-time algorithms [Reg09, GPV08]. To show the above theorem, we adapt the idea of using sparse secrets and invoking LHL (Left Over Hash lemma) sketched in [YZ16] to the case of LWE and show that a PKE scheme can be constructed assuming $2^{\frac{m}{\log \frac{m}{\alpha}}}$ hardness of LWE with an error rate of $\alpha = \frac{1}{\log \log r}$. The increase in the error rate of LWE problem is important because solving an LWE problem with higher error rate is equivalent to

given density

[5] Assuming LWE is $T(r)$ hard informally means assuming that any algorithm that runs in time less than $T(r)$ cannot solve LWE with given parameters

solving worst-case lattice problems such as gapSVP and SIVP with approximation factor $p(m)/\alpha$ for some polynomial p. And, lower the approximation factor for the worst-case lattice problems, the harder is to solve them. Therefore, it is ideal to assume hardness of LWE problem with higher α value.

Theorem 1.4 (PKE from k-SUM and LWE)**.** *For any security parameter $r \in \mathbb{N}$ and error rate $\alpha = \frac{1}{\omega(1)}$ there exists values $m = \log r \cdot \omega(1)$, $k = \omega(1)$, $q \geq 2\sqrt{m}\,k$ such that assuming the corresponding Learning With Error (LWE) problem with parameters r, m, k, α to be $2^{\frac{m}{\log \frac{m}{\alpha}}}$[6]-hard along with assuming k-SUM to be hard at density $\frac{1}{\mathsf{polylog}(r)}$ gives us a PKE scheme that satisfies weak* IND-CPA *security (Definition 4.1).*

This result is one of our main contributions. In this, we show that to increase the error factor to any value $\frac{1}{\omega(1)}$ that is greater than $\frac{1}{\log\log r}$, invoking LWE hardness alone is not sufficient and that we have to invoke the hardness of k-SUM at $\frac{1}{\mathsf{polylog}(r)}$ density. Hence, establishing the non-trivial contribution of k-SUM in increasing the error rate.

Both the above theorems are different parameter instantiations of Theorem 4.2, one can find the different parameter settings along with other analysis related to parameter setting in Section 4.2.

Lemma 1.5 (Theorem 4.3)**.** *For any security parameter $r \in \mathbb{N}$ and error rate $\alpha = \frac{1}{\omega(1)}$, there exists values m, $k = \omega(1)$, $q \geq 2\sqrt{m}\,k$ such that assuming the corresponding Learning With Error (LWE) problem with parameters r, m, k, α is 2^{m^c}-hard for any $c < 0.5$ along with assuming k-SUM to be hard at density $\frac{1}{\mathsf{polylog}(r)}$ gives us a PKE scheme that satisfies weak* IND-CPA *security (Definition 4.1).*

[6] Assuming LWE is $T(r)$ hard informally means assuming that any algorithm that runs in time less than $T(r)$ cannot solve LWE with given parameters

This result is important in the event of finding better algorithms for LWE[7]. We show that the security of our PKE construction can be based on assuming LWE with any super constant error rate to be 2^{m^c} hard for any $c < 0.5$ and k-SUM to be hard at density $= \frac{1}{\mathsf{polylog}(r)}$. To the best our knowledge this is the first PKE scheme to be based on the weaker assumption that any super constant approximation factor LWE that is only 2^{m^c} hard for $c < 0.5$.

New reductions and algorithms. As part of the above PKE construction, we define and use two variants of the k-SUM problem called the vector k-SUM problem and the targeted vector k-SUM problem. In the vector k-SUM problem, the elements are m-dimensional vectors from $\mathbb{Z}_q^m$ instead of being from $\mathbb{Z}_q$ as in the traditional k-SUM problem. In the targeted version of the (vector) k-SUM, instead of requiring the elements to sum to 0, we ask for the sum to be equal to a uniformly sampled value. This is thematically similar to how the Shortest Integer Solution (SIS) problem and the Inhomogenous SIS problem are related. We also provide various reductions establishing the hardness of these variants from the hardness of the k-SUM problem. More formally, we show the following:

Corollary 1.5.1 (Corollary 3.1.1)**.** *If planted search* kSUM *problem is hard at density* Δ*, then there is no algorithm that runs in time* $O\left(r^{k/2\,(1-\Omega(1))}\,k^{-m}\right)$ *and solves the decision* vector-kSUM *problem at density* Δ *with advantage* $o(1)$.

Corollary 1.5.2 (Corollary 3.3.1)**.** *If planted search* kSUM *is hard at density* Δ*, then there is no algorithm that runs in time* $O\left(r^{k/2\,(1-\Omega(1))}\,k^{-m}\right)$ *and solves the* targeted-vector-kSUM *problem at density* Δ *with advantage* $o(1)$.

We present a new algorithm for average-case k-XOR that is faster than known worst-case

[7]Current best known algorithm runs in time $2^{\frac{m}{\log\frac{m}{\alpha}}}$

algorithms at low densities in Corollary 3.4.1 i.e,

Theorem 1.6 (Algorithm for k-XOR, Theorem 3.4)**.** *For any $k \in \mathbb{N}$, there is an algorithm that runs in time $\widetilde{O}\left(r^k m^{3-k}\right)$ and solves the planted search k-XOR problem at any density with success probability* $1 - o(1)$.

Intuitively, given that we operate at a relatively low density, we exploit that with high probability, the planted linear relation turns out to be the only linear relation, which, in turn, can be easily found using Gaussian elimination. The new algorithm that we provide works by dividing the k-XOR instance (the matrix representing the k-XOR instance) into smaller instances (i.e., considering the sub-matrices) and then repeatedly using Gaussian elimination on the sub-matrices to find the original k-SUM solution.

Reducing public-key size using k-SUM. With an insight into interpreting the k-SUM family of problems as a computational variant of the famous Leftover Hash Lemma (LHL) [HILL99]. We notice that k-SUM can help not only weaken the required hardness from the "core" problem being used in the cryptographic construction but may also improve the overall efficiency of the construction. This is evident in the reduction in public key size of PKE with and without assuming k-SUM hardness. We show that

Public Key (pk) with k-SUM	Public Key (pk) without k-SUM
$\lvert\mathsf{pk}\rvert = r^{1/\log\log r}\,(2c\ \log r\ \log\log r)$	$\lvert\mathsf{pk}\rvert = r^2\,(2c\ \log r\ \log\log r)$

For further details on parameter analysis, please refer to Section 4.2.

Universally Composable Oblivious Transfer. We notice that the PKE scheme using LPN and k-XOR mentioned in [ASS+23] can be used in a black-box way (without any additional assumptions) to construct an extractable commitment scheme (Claim 4.10.1) which inturn implies a Universally Composable Oblivious Transfer (UC-OT) protocol due to the results in [DDN14]. More formally, we prove the following:

Theorem 1.7 (UC-OT from k-XOR and LPN, Theorem 4.11)**.** *There are values of noise rate $\eta = \Theta(1)$, $k = O(\log r)$ and $m = r^{o(1)}$, such that there exists a UC-OT protocol if the following condition is satisfied:*

- *Constant-noise LPN is 2^{m^c}-hard (for any constant $c > 0$) and planted search k-XOR is $r^{k/2-o(1)}$-hard at every density $\frac{1}{\mathrm{polylog}(r)}$ where m is the dimension of secret used in LPN.*

Intriguingly, the level of hardness needed from LPN in the above-said UC-OT protocol is not known to give a UC-OT protocol by itself. The best known constant-noise rate LPN hardness on which a UC-OT protocol has been based before this was $2^{m^{0.5}}$ [DDN14].

Somewhat Homomorphic Encryption. We notice that by taking a modified LPN problem where we replace the Bernoulli error with a k-XOR instance, we obtain the functionality (correctness) of a (Secret Key) Somewhat Homomorphic Encryption (Section 4.3). But, this construction turns out to be insecure. Informally, this is because when one gives out sufficiently many ciphertexts as part of the traditional IND-CPA security game, the linear relation present in the error of the modified LPN problem makes it susceptible to Gaussian elimination based attacks. However, a key observation is that all the linear attacks that involve Gaussian elimination require sufficiently many ciphertexts that correctly encrypt the message; let's say this number is some integer t_1. Therefore, we encrypt the wrong message as part of the scheme with a low probability such that this low probability will make sure that whenever an adversary takes t_1 many ciphertexts, at least one of them is erroneous while having a negligible impact on the decryption correctness. This thwarts any Gaussian elimination based attacks as that would require at least t_1 many correct encryptions. To capture this idea, we introduce a new assumption called probabilistic planted error LPN (PPE-LPN) (Definition 4.3) that roots its inspiration in the k-SUM problem. But, due to the modified error and erroneous encryption, our scheme ends up supporting a limited number of evaluations. More formally we show that

Lemma 1.8 (Theorems 4.8 and 4.9)**.** *For any $r \in \mathbb{N}$, $n = r$ and $m > \omega(1) \cdot \log r$ such that $k = \omega(1)$ is an odd integer and $\eta = \Omega(\frac{1}{(nm)})$ (denotes the rate of wrong encryption) assuming the hardness of probabilistic planted error LPN ($\mathsf{PPE} - \mathsf{LPN}(m, n, k, \eta)$) there exists a secret-key* SHE *which is IND-CPA secure and has the following properties:*

- *It can correctly decrypt any ciphertext that is generated homomorphically using at most $(mn)^{0.9}$ many evaluations (both addition and multiplication combined).*
- *Out of the $(mn)^{0.9}$ many evaluations, it can support constant number of multiplications and the size of the ciphertext generated is $(nm)^z$ where z is a constant denoting number of multiplication operations.*

Our SHE is interesting because many attempts to build Fully Homomorphic Encryption (FHE) from assumptions other than LWE have failed [8]. While even our scheme does not achieve fully homomorphic encryption, it can be looked at as a technical contribution towards a possible realisation of an LPN based (or any non-lattice problem based) FHE in a way similar to the contribution of [MGH08] to the lattice problem based FHE. Interestingly [MGH08] described a template for constructing Homomorphic Encryption schemes where the ciphertext grows exponentially with the multiplication depth but additions are supported without increasing the size. This is similar to the notion of SHE that we consider in this paper.

This brings us to the fact that the correctness and security properties by themselves do not make the SHE scheme interesting. An important property for an SHE is compactness (wrt the size of the ciphertext). According to [Hal17], a homomorphic encryption scheme is uninteresting if it cannot beat the trivial homomorphic in terms of size of ciphertext.

[8] For more details on these attempts refer to Section 1.2 and [Hal17]

A trivial (homomorphic) scheme is one that gives out all the ciphertexts and function description and expects the client to decrypt all the ciphertexts and evaluate the function on the decrypted messages. We show that our scheme is better than the trivial scheme in terms of compactness. We also note that our SHE scheme has the property that the ciphertext grows exponentially with the multiplications but additions are supported without increasing the size similar to the notion of limited homomorphism considered in [MGH08].

Collision Resistant Hash Function. Contrary to this work's main body, in the appendix, we assume the hardness of k-XOR in the dense regime and give a construction of a CRHF (Collision Resistant Hash Function).

As part of the analysis of our CPRF construction, we define another variant of the k-SUM problem called atmost vector-kSUM (avkSUM) which informally requires that i-SUM problem is hard for any $i \in [2, k]$ for a given value of k. We also provide justification for the hardness of this variant.

Theorem 1.9 (Theorem A.2)**.** *For any security parameter $r \in \mathbb{N}$, $k = \omega(1)$ and $m = \frac{k \log r}{3}$, assuming the hardness of atmost vector k-SUM (avkSUM) with parameters r, k, m, we show the existence of a CRHF.*

It is worth noting that the hash function itself is just $\mathsf{Hash}_{\mathbf{A}}(\mathbf{x}) := \mathbf{A} \cdot \mathbf{x}$ where the matrix $\mathbf{A}$ is sampled uniformly. This is the same as the usual SIS-based CRHF. It is just that we argue it's security using avkSUM at density $\Delta > 1$ instead of invoking the hardness of SIS problem. This ties up well with the initial motivation of ours to find a variety of hard problems to build cryptography on. Let's say some day we find an algorithm for the SIS problem, on that day we can still argue the security of CRHF used in primitives like hash encryption [DGHM17] using the hardness of avkSUM. However, unlike k-SUM in the sparse regime, this does not help in improving parameters of the CRHF (or) in reducing the hardness of LPN problem used in applications such as hash encryption.

1.2 RELATED WORK

The worst-case complexity of the k-SUM problem has been studied extensively in the field of fine-grained complexity due to its reductions to a large number of other interesting problems [GO95, BHP01, SEO03, BDP08, Pat10, AW14, KPP16, DSW18, Cha20, . . .]. We refer the reader to the survey by Williams [Wil18] for details. The complexity of the k-SUM problem in other computational models has also been studied, and it is known to have non-trivial decision trees [GS17, GP18], non-deterministic algorithms [CGI+16], and lower bounds in some of these models [Eri95, AC05]. Questions regarding data structures for it have also been studied [KP19, GGH+20, CL23].

Some conditional bounds for worst-case k-SUM are known in certain settings. For super-constant k, an algorithm that runs in time $r^{o(k)}$ would contradict the Exponential Time Hypothesis (ETH) [PW10]. Additionally, an algorithm for k-SUM with numbers in the range $\{0, \ldots, M-1\}$ that runs in time $M^{1-\Omega(1)}$ would contradict the Strong Exponential Time Hypothesis (SETH) [ABHS19].

Average-Case k-SUM. Average-case k-SUM and k-XOR in the dense regime have several applications in cryptanalysis and have been the subject of substantial work in the area, most involving better algorithms and applications [Wag02, MS12, NS15, Nan15, Din19, LS19, BDJ21].

More recently, different conditional lower bounds have been shown in this regime. Brakerski, Stephens-Davidowitz, and Vaikuntanathan [BSV21] show that Wagner's algorithm is near-optimal for k-SUM at large densities as k tends to infinity, using reductions from worst-case lattice problems. Dinur, Keller, and Klein [DKK21], as discussed above, show lower bounds at densities in $(1, 2)$ assuming the k-SUM conjecture at density 1. Dalirrooyfard, Lincoln, and Williams [DLW20] show the average-case hardness of counting solutions in a "factored" version of k-SUM assuming SETH. They also show search-to-decision reductions for the average-case Zero-k-Clique problem.

The study of average-case fine-grained complexity, in general, has proliferated in the past few years [BRSV17, DLW20]. Of particular relevance here is a line of work on worst-case to average-case reductions for counting k-cliques, which focuses on reducing from and to the same problem [GR18, BBB19]. The general paradigm of looking for small hidden solutions in random instances is common in problems studied in statistical inference, such as planted clique, Sparse PCA, etc. [Jer92, BR13b, BR13a, GZ19]. Worst-case versions of these problems have also been subjects of interest in fine-grained complexity [Wil18, GV21].

Fine-Grained Cryptography. The question of constructing cryptographic primitives with fine-grained security guarantees assuming fine-grained hardness conjectures has been studied alongside average-case fine-grained complexity [BRSV18, LLW19, BC22]. LaVigne, Lincoln, and Williams [LLW19] use an assumption about the hardness of decision k-SUM to construct a fine-grained One-Way Function. They also construct fine-grained Public-Key Encryption (with quadratic security) assuming the average-case hardness of the Zero-k-Clique problem. Juels and Peinado [JP00] similarly constructed One-Way Functions from the conjectured hardness of planted clique for certain parameters.

Structured problems where a hidden solution can be planted have also been used to construct cryptography in [ABW10, LLW19]. An immediately relevant illustration of this may be seen in the case of the subset sum problem, which is the unparametrized version of k-SUM where the size of the solution is not restricted. The average-case hardness of the planted subset sum problem at very low densities has been used to construct Public Key Encryption by Lyubashevsky, Palacio, and Segev [LPS10]. However, it is worth noting that the PKE construction based on LPN and k-XOR from [ASS+23] is not a simple modification of this construction. In fact, the appropriate adaptation of their construction to k-XOR would be insecure[9].

Homomorphic Encryption. The evolution of Homomorphic Encryption can be charted to the early 1980s where non compact Fully Homomorphic Encryption (FHE) (where the size of the ciphertext grows with the complexity of the evaluated function) were known to have existed [Yao82]. At the same time the compact additively homomorphic (or) multiplicatively homomorphic schemes that support only addition or only multiplication on encrypted data also existed [GM84, Elg85]. However, it wasn't until [BGN05, GHV10] that there existed a compact scheme that supported unbounded number of additions and one multiplication. The security of [BGN05] and [GHV10] schemes were based on group theoretic and LWE hardness assumptions respectively. Similarly schemes like [IP07, MGH08] exhibited limited homomorphism. Interestingly [MGH08] described a template for constructing Homomorphic Encryption schemes where the ciphertext grows exponentially with the multiplication depth but additions are supported without increasing the size. This is almost identical to the notion of SHE that we consider in this paper. Then [Gen09] came up with the ground breaking result of FHE and this was followed by many improvements and different approaches. Few of the important ones are [BGV11] and [GSW13].

It is worth noting that many attempts to build FHE from assumption other than LWE have failed. Since the problem of learning-parity-with-noise (LPN) is similar to LWE, one would have expected an FHE based on the LPN problem. However, there are no known constructions of FHE based on LPN problem. One such attempt was made by [BL11], however it was shown to be insecure by [GOT12]. On top of it, the result by [Bra12] informally says that the decryption procedure of an FHE scheme from LPN must be more complicated than a simple inner product of some LPN based schemes such as [Ale03b].

[9] Briefly, the construction by [LPS10] relies on the hardness of subset-sum (or possibly k-SUM) at a density where the number of bits in each element is roughly equal to the number of elements in an instance. At this very low density, k-XOR (unlike subset-sum or k-SUM) can be easily solved using Gaussian elimination (see Section 3.3).

Therefore, as mentioned above, we hope our work contributes to an LPN based FHE in a way similar to the contribution of [MGH08] to LWE based FHE. For many more interesting discussions about evolution of homomorphic encryption and more, we refer the reader to [Hal17].

CHAPTER 2

PRELIMINARIES

In this chapter, we provide the necessary preliminaries used in the rest of the work. Additional preliminaries (or) definitions are given in their respective chapters.

Notation. We denote by $\log x$ the base-2 logarithm of x. For a real number $x \in \mathbb{R}$, we let $\lfloor x \rfloor$ denote the largest integer not greater than x, $\lceil x \rceil$ denote the smallest integer greater than x, and $\lfloor x \rceil$ denote the integer closest to x, with ties broken upward. We use $[a, b]$ to denote the set of integers $\{k \in \mathbb{N} \mid a \le k \le b\}$. We use $[n]$ to denote the set $[1, n]$. We say a function $f(r)$ is *negligible* if it is $O(r^{-c})$ for all $c > 0$, and we use $\mathrm{negl}(r)$ to denote a negligible function of r. A function that is not negligible is denoted by non-$\mathrm{negl}(r)$. In asymptotic notation, the use of a tilde – $\tilde{O}(\cdot), \tilde{\Omega}(\cdot)$ – indicates that poly-logarithmic factors in the main parameter are ignored.

Vectors. We use bold lowercase letters ($\mathbf{x}$) and bold uppercase letters ($\mathbf{M}$) to denote vectors and matrices respectively. By default, vectors are column vectors. For any vector $\mathbf{x}$ of length ℓ, we let x_i denote the i-th coordinate of $\mathbf{x}$, for $i \in [\ell]$. We use $\mathbf{x}^T$ (resp. $\mathbf{A}^T$) to represent the transpose of a vector (resp. a matrix). If $\mathbf{u}, \mathbf{v} \in \mathbb{F}_2^m$ then we denote coordinate-wise XOR of $\mathbf{u}$ and $\mathbf{v}$ (i.e. $[\mathbf{u} \oplus \mathbf{v}]_i = \mathbf{u}_i \oplus \mathbf{v}_i$ for $i = 1 \ldots m$) by $\mathbf{u} \oplus \mathbf{v}$. For a vector $\mathbf{u} \in \mathbb{R}^m$ and $q \ge 1$, we denote by $\|\mathbf{u}\|_q$ the ℓ_q-norm of $\mathbf{u}$, defined as:

$$\|\mathbf{u}\|_q := \left(\sum_{i=1}^{m} \mathbf{u}_i^q \right)^{1/q}$$

We denote by $\|\mathbf{u}\|_0$ the number of non-zero entries in $\mathbf{u}$, i.e. $\|\mathbf{u}\|_0 = \sum_{i=1}^{m} \mathbf{u}_i$.

Distributions. With some exceptions, we will use uppercase letters to denote random variables and distributions. For a distribution D, we use the notation $x \leftarrow D$ to denote the process of sampling a random value according to the distribution. For a finite set S, we use the notation $s \xleftarrow{\$} S$ to denote the process of choosing s uniformly at random from S. When we say $\forall\, x \in X$, where X is a distribution or a randomized algorithm, it means $\forall\, x$ in support of X.

The *statistical distance* between two random variables X and Y with supports $\mathcal{X}$, $\mathcal{Y}$ respectively is denoted by $\mathsf{SD}(X;Y)$ and is defined as follows:

$$\mathsf{SD}\,(X;Y) := \frac{1}{2} \sum_{u \in \mathcal{X} \cup \mathcal{Y}} |Pr[X = u] - Pr[Y = u]|$$

Distributions D_1 and D_2 over $\{0,1\}^n$ are *computationally indistinguishable* (denoted $D_1 \approx_c D_2$) if no $\mathsf{poly}(n)$-time algorithm can distinguish between them with advantage non-negligible in n. They are *statistically indistinguishable* ($D_1 \approx_s D_2$) if the statistical distance between them is negligible in n. In addition, we write $\mathcal{D}_1 \equiv \mathcal{D}_2$ when $\mathcal{D}_1$ and $\mathcal{D}_2$ are the same distribution.

We denote by Ber_η a Bernoulli random variable on $\mathbb{F}_2$ with parameter η, and by Ber_η^n a vector on $\mathbb{F}_2^n$ where each entry is i.i.d. Ber_η.

Concentration Bounds.

Lemma 2.1 (Chernoff Bound)**.** *Let $X_1, X_2, \ldots, X_n$ be i.i.d random variables on $\{0,1\}$, and let $X = \sum_{i=1}^n X_i$. Then for every $\varepsilon > 0$,*

$$\Pr[X > (1+\varepsilon)\,\mathbb{E}[X]] < \mathrm{e}^{-\frac{\varepsilon^2\,\mathsf{E}[X]}{2}},$$
$$\text{and,} \quad \Pr[X < (1-\varepsilon)\,\mathsf{E}[X]] < \mathrm{e}^{-\frac{\varepsilon^2\,\mathsf{E}[X]}{2}}.$$

Similarly, the following also holds,

$$\Pr\left[\frac{1}{n}X > \frac{1}{n}\mathbb{E}[X] + \varepsilon\right] < \mathrm{e}^{-2\varepsilon^2 n},$$

$$\text{and,} \quad \Pr\left[\frac{1}{n}X < \frac{1}{n}\mathsf{E}[X] - \varepsilon\right] < \mathrm{e}^{-2\varepsilon^2 n}.$$

Extractors.

Lemma 2.2. *(Leftover Hash Lemma)([HILL99, GKPV10]) Let $(X, Z) \in \mathcal{X} \times \mathcal{Z}$ be any joint random variable over $\mathbb{Z}_q^n$ with min-entropy $H_\infty(X|Z) \geq k$. For any $\epsilon > 0$ and $m \leq \frac{k - 2\log(\frac{1}{\epsilon}) - O(1)}{\log q}$, the joint distribution of $Z \times (\mathbf{C}, \mathbf{C} \cdot \mathbf{s})$ where $\mathbf{C} \xleftarrow{\$} \mathbb{Z}_q^{m \times n}$ is uniformly random and $\mathbf{s} \in \mathcal{X}$ is ϵ-close to the uniform distribution over $Z \times (\mathbb{Z}_q^{m \times n}, \mathbb{Z}_q^m)$.*

Discrete Gaussian. Regev [Reg09] defined a natural distribution over lattices called the discrete Gaussian distribution, parametrized by a scalar $\alpha > 0$. We additionally need a bound on samples drawn from the discrete Gaussian distribution.

Lemma 2.3. *[Lyu12] Let χ_σ be the discrete Gaussian distribution on $\mathbb{Z}$ with parameter σ. Then, for any $k > 0$, $\Pr[|z| > k\sigma; z \leftarrow \chi_\sigma] \leq 2e^{(-k^2/2)}$.*

2.1 LWE AND LPN

Here, we introduce the decision versions of the Learning With Errors (LWE) and the Learning Parity with Noise (LPN) problems and mention some of the relevant known results about them.

Definition 2.1 (LWE Problem). Let r be the security parameter, let $m = m(r)$, $n = n(r)$ and $q = q(r) > 2$ be integers, and $\chi = \chi(r)$ be a distribution over $\mathbb{Z}_q$. The $\mathsf{LWE}(m, n, q, \chi)$ problem over $\mathbb{Z}_q$ is to distinguish between the following distributions:

- $(\mathbf{A}, \mathbf{s}^T\mathbf{A} + \mathbf{e}^T)$, where $\mathbf{A} \leftarrow \mathbb{Z}_q^{m \times n}$, $\mathbf{s} \leftarrow \mathbb{Z}_q^m$, and $\mathbf{e} \leftarrow \chi^n$
- $(\mathbf{A}, \mathbf{v}^T)$ where $\mathbf{A} \leftarrow \mathbb{Z}_q^{m \times n}$, and $\mathbf{v} \leftarrow \mathbb{Z}_q^n$

We say that an algorithm $\mathcal{A}$ solves the LWE problem with advantage ϵ if,

$$\left|\Pr[\mathcal{A}(\mathbf{A}, \mathbf{s}^T\mathbf{A} + \mathbf{e}^T) = 1] - \Pr[\mathcal{A}(\mathbf{A}, \mathbf{v}^T) = 1]\right| \geq \epsilon,$$

Where the randomness is taken over the instance, and the random coins are used by $\mathcal{A}$. We say that the $\mathsf{LWE}(m, n, q, \chi)$ problem is T*-hard* if no adversary $\mathcal{A}$ running in time $T(r)$ can distinguish between the above distributions with advantage greater than $1/T(r)$. It is known ([Reg09, BLP+13]) that if we set χ to be the discrete Gaussian distribution with parameter αq, the $\mathsf{LWE}(m, n, q, \chi)$ problem is as hard as solving worst-case lattice problems such as gapSVP and SIVP with approximation factor $p(m)/\alpha$ for some polynomial p.

Theorem 2.4. *[Reg09, Pei09, BLP+13] For any $n = \mathsf{poly}(m)$, any modulus $q \leq 2^{\mathsf{poly}(m)}$, and any (discretized) gaussian error distribution χ of parameter $\alpha q \geq 2\sqrt{m}$ where $0 < \alpha < 1$, solving the $\mathsf{LWE}(m, n, q, \chi)$ problem is at least as hard as solving gapSVP_γ on arbitrary m-dimensional lattices, for some $\gamma = \tilde{O}(\frac{m}{\alpha})$*

Since the best-known algorithms for 2^k-approximation of gapSVP and SIVP run in time $2^{\tilde{O}(m/k)}$[Pei15], Theorem 2.4 implies that the best known algorithms that solve LWE run in time $\Omega(2^{\frac{m}{\log \frac{m}{\alpha}}})$.

The Learning Parity with Noise problem may be seen as an analogue of LWE with a different notion of noise. We define it formally as follows.

Definition 2.2 (Search LPN Problem). An algorithm $\mathcal{A}$ is said to solve the *search Learning Parity with Noise (LPN) problem with noise rate* η with probability ϵ if, given $\langle \mathbf{X}, \mathbf{X}\,\mathbf{s} + \mathbf{e} \rangle$ where $\mathbf{X} \leftarrow \mathbb{F}_2^{r \times m}$, $\mathbf{s} \leftarrow \mathbb{F}_2^m$, and $\mathbf{e} \leftarrow \mathrm{Ber}_\eta^r$, it outputs $\mathbf{s}$ with probability at least ϵ, where the randomness is taken over the instance and the random coins used by the algorithm $\mathcal{A}$.

The best-known algorithm for search LPN when $\eta = \Theta(1)$ is by Blum, Kalai, and Wasserman [BKW03] that for an m-dimensional secret runs in subexponential time

$2^{O(m/\log(m))}$. There is another algorithm by Esser, Kübler, and May [EKM17] that runs in time $2^{O(\eta m)}$ and thus outperforms the BKW algorithm when the noise rate is small. We also consider the decision LPN problem that we formally define as follows.

Definition 2.3 (Decision LPN Problem). The *Decision Learning Parity with Noise (LPN) problem with noise rate* $\eta \in (0,1)$ is to distinguish between the following distributions:

1. $\langle \mathbf{X}, \mathbf{X}\,\mathbf{s} + \mathbf{e}\rangle$, where $\mathbf{X} \leftarrow \mathbb{F}_2^{r\times m}$, $\mathbf{s} \leftarrow \mathbb{F}_2^m$, and $\mathbf{e} \leftarrow \mathrm{Ber}_\eta^r$.
2. $\langle \mathbf{X}, \mathbf{y}\rangle$, where $\mathbf{X} \leftarrow \mathbb{F}_2^{r\times m}$, and $\mathbf{y} \leftarrow \mathbb{F}_2^r$.

We say that an algorithm $\mathcal{A}$ solves the decision LPN problem with advantage ϵ if,

$$\Big|\Pr[\mathcal{A}(\mathbf{X}, \mathbf{X}\,\mathbf{s} + \mathbf{e}) = 1] - \Pr[\mathcal{A}(\mathbf{X}, \mathbf{y}) = 1]\Big| \geq \epsilon,$$

Where the randomness is taken over the instance, and the random coins are used by $\mathcal{A}$. The search and decision LPN problems are known to be polynomially equivalent, as shown in [KSS10]. We restate their result as follows.

Lemma 2.5 (Search-to-Decision LPN [KSS10])**.** *If there is an algorithm that runs in time T and solves the decision LPN problem with advantage ϵ, then there is an algorithm that runs in time $O\left(\frac{T\,m\log m}{\epsilon^2}\right)$ and solves the search LPN problem with probability $\epsilon/4$.*

Definition 2.4 (Hardness of LPN). We say that *η-noise search LPN is $T(m)$-hard* if any algorithm A that runs in time at most $T(m)$ has success probability at most $o(1/T(m))$ in solving the search LPN problem for secrets of size m with noise rate $\Omega(\eta)$.

We can leverage the equivalence of Theorem 2.5 to translate the hardness of LPN into a bound on the advantage of an algorithm that solves decision LPN.

Corollary 2.5.1. *If η-noise Search LPN is $(T(m)^3 \cdot m\log m)$-hard, then there is no algorithm that runs in time $O(T(m))$ and solves the decision LPN problem with advantage*

$\Omega(1/T(m))$.

2.2 K-SUM AND RELATED DEFINITIONS

In this section, we introduce the kSUM and kXOR problems and discuss some of their known results that we will use throughout this work. By default, these terms will refer to the *decision* versions of these problems, and we will qualify the (planted) search versions whenever they are discussed.

Definition 2.5 (kSUM problem). Let $r, k = k(r), q = q(r)$ be integers. The $\mathsf{kSUM}(r, k, q)$ problem is to distinguish between the following distributions:

- $(a_1, a_2, \ldots, a_r)$, where each a_i is chosen uniformly from $\mathbb{Z}_q$.
- $(a_1, a_2, \ldots, a_r)$, where each a_i is chosen uniformly from $\mathbb{Z}_q$, and then for a uniformly chosen set $S \subseteq [r]$ of size k, we set $a_j = -\sum_{i \in S \setminus \{j\}} a_i$, where $j = \min_{i \in S} i$.

That is, the task is to distinguish between a set of n random numbers from $\mathbb{Z}_q$ and one with a planted set of k numbers that sum to 0. We shall call such a set of k numbers a *solution*. We will refer to the first distribution above as the *non-planted* distribution and the second (with the planted solution) as the *planted* distribution.

The feasibility of this problem depends on the relative values of r, k, and q. As an aid to discussion, we use the following quantity called the *density* of the $\mathsf{kSUM}(r, k, q)$ problem:

$$\Delta = \frac{k \log r}{\log q}$$

Roughly, when $\Delta = 1$, a uniformly random set of r elements from $\mathbb{Z}_q$ will contain a solution with some constant probability. As Δ increases beyond 1, there will exist many such solutions; further, the planted and non-planted distributions become statistically close, and the kSUM problem is not feasible to solve with a large advantage.

For $\Delta < 1$, the distributions are statistically far, and near-optimal distinguishing advantage can be achieved by algorithms that run in time $\widetilde{O}(r^{\lceil k/2 \rceil})$. These are essentially the algorithms that solve the *search* version of the kSUM problem in the *worst case* [BDP08, GP18, Cha20]. Except at very small values of Δ (roughly $\widetilde{O}(1/r^2)$ [ASS+23]), no better algorithms are known for the kSUM problem in this regime. It has been conjectured that, at least at density $\Delta = 1$, this is the best algorithm we can have for this problem.

Conjecture 2.6 ([Pet15, LLW19, DKK21]). *For r, $k = k(r)$, and $q = q(r)$ such that $k \log r \approx \log q$, any algorithm that runs in time $r^{\lceil k/2 \rceil (1-\Omega(1))}$ solves* $\mathsf{kSUM}(r, k, q)$ *with advantage at most $o(1)$.*

Looking ahead, in our constructions, we will use the assumption that kSUM and related problems are nearly as hard even at some densities $\Delta = \Theta(1/\mathsf{polylog}(r))$. So far, we do not know any algorithms at these densities that are more efficient than the worst-case ones mentioned above; a more detailed analysis of the hardness of the k-SUM family of problems can be seen in [ASS+23].

Search and Decision variants. Next, we provide search and decision variants of the k-SUM and k-XOR problem and the relation between these.

Definition 2.6 (Search-k-SUM problem). Let $r, k = k(r), q = q(r)$ be integers and let $\Delta = \left\lceil \frac{k \log r}{\log q} \right\rceil$. An algorithm $\mathcal{A}$ is said to solve Search-$\mathsf{kSUM}(r, k, q)$ problem with success probability ϵ if, on input an instance $\mathbf{a}$ of size r it outputs an $S = \mathcal{A}(\mathbf{a}) \subseteq [r]$ with $|S| = k$ such that:

$$\Pr_{S \leftarrow \mathcal{A}(\mathbf{a})} \left[\sum_{i \in S} a_i = 0 \right] \geq \epsilon$$

where $\mathbf{a} \xleftarrow{\$} \mathbb{Z}_q^r$ where a_i denotes the i^{th} element of the vector $\mathbf{a}$.

Lemma 2.7 (Search-to-Decision reduction for kSUM [ABRV23]). *Let $n, k = k(n), q = q(n)$ be integers. Suppose there is an algorithm that runs in time T and solves* $\mathsf{kSUM}(n, k, q)$ *with advantage $\Omega(1)$, where $T = \Omega(n(k \log^2 n + \log q))$. Then, there is an algorithm that runs in time $\widetilde{O}(k^2 T)$ and solves Search-*$\mathsf{kSUM}(n, k, q)$ *with success probability $\Omega(1)$.*

Theorem 2.8 (Search-to-Decision Reduction, implied by [IN89]). *For $k \in \mathbb{N}$ and $\Delta < 1$, suppose there is an algorithm that runs in time $T(r)$, and solves the* decision *k-SUM problem i.e,* $\mathsf{kSUM}(r, k, q)$ *with success probability $(1/2 + \epsilon)$. Then, there is an algorithm that runs in time $O(T(r) \cdot r \log(r) \cdot (k/\epsilon)^4)$, and solves the* planted search *k-SUM problem i.e, Search-*$\mathsf{kSUM}(r, k, q)$ *with success probability at least $3/4$.*

Definition 2.7 (kXOR problem). Let $r, k = k(r), m = m(r)$ be integers. The $\mathsf{kXOR}(r, m, k)$ problem is to distinguish between the following distributions:

- $(a_1, a_2, \ldots, a_r)$, where each a_i is chosen uniformly from $\mathbb{Z}_2^m$.

- $(a_1, a_2, \ldots, a_r)$, where each a_i is chosen uniformly from $\mathbb{Z}_2^m$, and then for a uniformly chosen set $S \subseteq [r]$ of size k, we set $a_j = \bigoplus_{i \in S \setminus \{j\}} a_i$, where $j = \min_{i \in S} i$.

That is, the task is to distinguish between a set of n random m-dimensional boolean vectors from $\mathbb{Z}_2^m$ and one with a planted set of k boolean vectors that sum to 0. We shall call such a set of k boolean m-dimensional vectors a *solution*. We will refer to the first distribution above as the *non-planted* distribution and the second (with the planted solution) as the *planted* distribution. Analogous to the kSUM problem we will denote the density as $\Delta = \frac{k \log n}{\log 2^m}$.

Definition 2.8 (Search-k-XOR problem). Let $r, m = m(r), k = k(r)$ be integers. and let $\Delta = \left\lceil \frac{k \log r}{m} \right\rceil$. An algorithm $\mathcal{A}$ is said to solve Search-$\mathsf{kXOR}(r, m, k)$ problem with success probability ϵ if, on input an instance $\mathbf{A}$ of size r it outputs an $S = \mathcal{A}(\mathbf{A}) \subseteq [r]$

with $|S| = k$ such that:

$$\Pr_{S \leftarrow \mathcal{A}(\mathbf{A})}\left[\bigoplus_{i \in S} \mathbf{a}_i = 0\right] \geq \epsilon$$

where $\mathbf{A} \xleftarrow{\$} \mathbb{Z}_2^{m \times r}$ where $\mathbf{a}_i$ is the i^{th} column of $\mathbf{A}$.

Lemma 2.9 (Search-to-Decision reduction for kXOR [ABRV23]). *Let $r, k = k(r), m = m(r)$ be integers. Suppose there is an algorithm that runs in time T and solves* $\mathsf{kXOR}(r, m, k)$ *with advantage $\Omega(1)$, where $T = \Omega(r(k \log^2 r + m))$. Then, there is an algorithm that runs in time $\widetilde{O}(k^2 T)$ and solves Search-*$\mathsf{kXOR}(r, k, q)$ *with success probability $\Omega(1)$.*

Theorem 2.10 (Search-to-Decision Reduction, implied by [IN89]). *For $k \in \mathbb{N}$ and $\Delta < 1$, suppose there is an algorithm that runs in time $T(r)$, and solves the* decision *k-SUM problem i.e,* $\mathsf{kXOR}(r, m, k)$ *with success probability $(1/2 + \epsilon)$. Then, there is an algorithm that runs in time $O(T(r) \cdot r \log(r) \cdot (k/\epsilon)^4)$, and solves the* planted search *k-SUM problem i.e, Search-*$\mathsf{kXOR}(r, m, k)$ *with success probability at least $3/4$.*

Definition 2.9 (Hardness of Search k-XOR problem). We say that (planted) search *k-XOR is $T(r)$-hard at density* Δ if any algorithm that runs in time at most $T(r)$ has a success probability at most $(1 - \Omega(1/\log r))$ in solving the planted search k-XOR problem at density Δ.

Lemma 2.11 ([ASS+23]). *If the planted search* kXOR *problem at density $\Omega\left(\frac{1}{\mathsf{polylog}(r)}\right) \leq \Delta \leq 1$ is hard to solve with probability $1 - o\left(\frac{1}{\log r}\right)$ in time $\widetilde{O}(T)$, it is also hard to solve with probability $\gamma = \Omega\left(\frac{1}{\mathsf{polylog}(r)}\right)$ in time $\widetilde{O}(T \cdot (\gamma^2/k)^k)$.*

Corollary 2.11.1. *For any $k = o(\log r/\log\log r)$, if planted search k-XOR is $T(r)$-hard at density Δ, then any algorithm that runs in time $\left(T(r)/r^2\right)$ solves the decision-k-XOR problem at density Δ with advantage at most $o(1)$.*

CHAPTER 3

NEW RESULTS ABOUT K-SUM AND COUSINS

In this chapter, we introduce new variants of kSUM called vector-kSUM and targeted-vector-kSUM and show various results involving and interlinking them. We follow this with a faster algorithm to solve the k-XOR problem (a variant of k-SUM).

3.1 VECTOR K-SUM

In this section, we will first define a variant of k-SUM called the vector-k-SUM and then proceed to show via a reduction that the hardness of decision vector k-SUM may be based on the hardness of decision k-SUM problem.

Definition 3.1 (Vector-k-SUM problem). Let $n, m = m(n), k = k(n), q = q(n)$ be integers and let $\Delta = \left\lceil \frac{k \log n}{m \log q} \right\rceil$. An algorithm $\mathcal{A}$ is said to solve the vector-kSUM(n, m, k, q) problem with success probability ϵ if, on input an instance $\mathbf{A}$ of size n it outputs an $S = \mathcal{A}(\mathbf{A}) \subseteq [n]$ with $2 \leq |S| \leq k$ such that:

$$\Pr_{S \leftarrow \mathcal{A}(\mathbf{A})} \left[\sum_{i \in S} \mathbf{a}_i = 0 \right] \geq \epsilon$$

where $\mathbf{A} \xleftarrow{\$} \mathbb{Z}_q^{m \times n}$ and $\mathbf{a}_i$ denotes the i^{th} column of $\mathbf{A}$.

3.1.1 Reduction from k-SUM to Vector k-SUM

Lemma 3.1. *If there is an algorithm that solves decision* vector-kSUM *at density Δ in time $T(r)$ with advantage ϵ, then there is an algorithm that solves decision* kSUM *at density Δ in time $O(T(r) \cdot k^m)$ with advantage ϵ.*

Proof. Assume that we have oracle access to an adversary $\mathcal{A}$ that solves decision vector-kSUM. It follows from the definition of decision vector-kSUM that $\mathcal{A}$ takes as input r vectors of length m, each of whose elements are members of $\mathbb{Z}_q$.

For the sake of simplicity, we assume that we are trying to solve decision kSUM modulo q^m and that q is a prime. It is easy to verify that a density Δ instance consists of r elements.

Our first step involves expressing each element in base q. This transforms an element of $\mathbb{Z}_{q^m}$ into a vector of size m with each element belonging to $\mathbb{Z}_q$. Observe that if k elements are added up to zero, each element of the sum of their image vectors must be within $[-(k-1), 0] \mod q$ since the carry at each position must be less than the number of summands. This suggests the following algorithm.

Algorithm $\mathcal{B}_1(\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_r)$

1. Create matrix $\mathbf{Y}$ of size $m \times r$ where $\mathbf{Y}[i][j]$ is the ith digit of the base q representation of $\mathbf{x}_j$.
2. For each vector $\mathbf{v} \in [0, k-1]^m \mod q$:
   - 2.1. Create a new matrix $\mathbf{Y}_\mathbf{v}$ where $\mathbf{Y}_\mathbf{v}[i][j] = \mathbf{Y}[i][j] + \mathbf{v}_i/k$
   - 2.2. Call $\mathcal{A}$ on $\mathbf{Y}_\mathbf{v}$. Say it returns b'
   - 2.3. If $b' = 1$ return 1
3. Return 0

Let us assume that the input $\mathbf{X}$ is a planted instance of decision kSUM at density Δ. Let S be the set of planted indices. Observe that there exists some particular $v \in [0, k-1]^m \mod q$ such that $\sum_{j \in S} \mathbf{Y}[i][j] + \mathbf{v}[i] = 0 \mod q$ for all $i \in [m]$ ($\mathbf{v}$ can be thought of as the carry vector). Since $|S| = k$, the above equation implies $\sum_{j \in S} \mathbf{Y}_\mathbf{v}[i][j] = 0 \mod q$ $\forall i \in [m]$. Furthermore, it is easy to see that $\mathbf{Y}_\mathbf{v}$ is sampled from the planted distribution on $\mathbb{Z}_q^m$. Let us further assume that the input $\mathbf{X}$ is a non-planted instance of decision kSUM at density Δ. This means that all the elements $\{\mathbf{x}_i\}_{i \in r]}$ are sampled uniformly at random $\implies$ $\mathbf{Y_v}$ is identical to being sampled uniformly.

By the above, we have shown that the distribution from which $\mathbf{Y_v}$ is sampled is identical to the distribution from which an instance of decision kSUM is sampled in both planted and non-planted cases, respectively. Hence, if $\mathcal{A}$ solves decision vector-kSUM with advantage ϵ then $\mathcal{B}_1$ solves decision kSUM with success probability ϵ.

The runtime of this algorithm is k^m times the runtime of $\mathcal{A}$ which simplifies to $O(k^m T)$. ■

Note: At low densities ($\Delta \leq \frac{1}{r^{0.5+\epsilon} \cdot \log q}$) where $\epsilon \geq \frac{3}{k-3}$, decision vector-kSUM can be solved in time $o(r^{\lceil \frac{k}{2} \rceil})$ by the algorithm described in Lemma 3.4.

Corollary 3.1.1. *If planted search* kSUM *problem is hard at density* Δ*, then there is no algorithm that runs in time* $O\left(r^{k/2\,(1-\Omega(1))}\, k^{-m}\right)$ *and solves the decision* vector-kSUM *problem at density* Δ *with advantage* $o(1)$.

Note that this follows from the conjectured hardness of the planted k-SUM problem, Theorem 2.10, and the above reduction.

3.2 TARGETED VECTOR K-SUM

In this section, we will define a variant of k-SUM called the targeted vector-k-SUM and then proceed to show via a reduction that the hardness of decision vector k-SUM may be based on the hardness of decision vector-k-SUM.

Definition 3.2 (Targeted vector k-SUM). For any $r \in \mathbb{N}$, $k \in \mathbb{N}$, $q \in \mathbb{N}$, $m \in \mathbb{N}$, an algorithm $\mathcal{A}$ is said to solve the targeted-vector-kSUM(r, k, q, m) with success probability ϵ if for both $b \in \{0, 1\}$,

$$\Pr_{\mathbf{X} \sim \mathsf{D}_b^{(r)}} [\mathcal{A}(\mathbf{X}) = b] \geq \epsilon.$$

If $\epsilon = 1 - o(1)$, we simply say that $\mathcal{A}$ solves the targeted-vector-kSUM.

where the distributions are defined below:

Distribution $\mathsf{D}_0^{(r)}$

1. Sample r group elements $\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_r$ i.i.d. uniformly at random from $\mathbb{Z}_q^m$
2. Return $(\mathbf{x}_1, \mathbf{X})$

Distribution $\mathsf{D}_1^{(r)}$

1. Sample $r-1$ group elements $\mathbf{x}_2, \ldots, \mathbf{x}_r$ i.i.d. uniformly at random from $\mathbb{Z}_q^m$
2. Choose a random set $S \subseteq [2, r]$ with $|S| = k-1$.
3. Compute $\mathbf{x}_1 := \sum_{j \in S} \mathbf{x}_j$
4. Return $(\mathbf{x}_1,\ \mathbf{X})$

Note that targeted-vector-kSUM, as defined above, is a decision problem. We do not prepend the word 'decision' to it because we only use the decision version of targeted-vector-kSUM in this work. Moreover, we define the density in this case equivalently as $\Delta = \frac{k \log r}{m \log q}$.

Conjecture 3.2. *For any $n, k = k(n), q = q(n), m = m(n)$, any adversary that runs in time $n^{\lceil k/2 \rceil (1-\Omega(1))-1}$ solves* targeted-kSUM(n, k, q, m) *with advantage at most $o(1)$.*

The above conjecture is justified in Lemma 3.3.

3.2.1 Reduction from Vector k-SUM to Targeted Vector k-SUM

Lemma 3.3. *If there is an algorithm that solves the targeted vector k-SUM problem at density Δ in time T with advantage ϵ, then there exists an algorithm that solves decision vector* kSUM *at density Δ in expected time $r\Theta(T)$ with advantage ϵ.*

Proof. Below, we give a reduction (algorithm $\mathcal{B}_2$) from targeted-vector-kSUM to vector-kSUM.

Algorithm $\mathcal{B}_2(\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_r)$

1. Repeat at most r times:

   a) Pick a permutation $\pi : [r] \to [r]$ uniformly. Let $\mathbf{X}' := [\mathbf{x}_{\pi(1)}, \ldots, \mathbf{x}_{\pi(r)}]$.

   b) Call $\mathcal{A}$ on the vector $\mathbf{X}' := [\mathbf{x}'_2, \ldots, \mathbf{x}'_r]$ and the target $\mathbf{x}'_1 := \mathbf{x}_{\pi(1)}$. Say it returns value b.

   c) If $b = 1$ then return 1 (Indicating a planted set)

2. Return 0

We can re-interpret the decision vector-kSUM problem defined at density Δ as follows:

1. Pick the (planted) index $i \xleftarrow{\$} [r - k + 1]$.
2. Pick a set S uniformly such that from $S \subseteq [r] \setminus [i]$ and $|S| = k - 1$.
3. Pick $a_1, \ldots a_r$ for all indices uniformly at random except i.
4. Let $\mathbf{x}_i = -\sum_{j \in S} \mathbf{x}_j$ (or) $\mathbf{x}_i \xleftarrow{\$} \mathbb{Z}_q^m$
5. Output: $[\mathbf{x}_1, \ldots, \mathbf{x}_r]$.

The targeted-vector-kSUM problem at density Δ can be re-interpreted as follows:

1. Pick a set S uniformly such that from $S \subseteq [2, r]$ and $|S| = k - 1$.
2. Pick $\mathbf{x}_2, \ldots \mathbf{x}_r$ for all indices uniformly.
3. Let $\mathbf{x}_1 = \sum_{j \in S} \mathbf{x}_j$ (or) $\mathbf{x}_1 \xleftarrow{\$} \mathbb{Z}_q^m$
4. Output: $([\mathbf{x}_2, \ldots, \mathbf{x}_n], \mathbf{x}_1)$.

Notice that the distributions of decision vector-kSUM and targeted-vector-kSUM are equivalent at density Δ, conditioned on the fact that (planted) index i sampled in vector-kSUM is equal to 1. By re-permuting the indices after every iteration in our reduction, we ensure that the planted index gets permuted to the location 1 (this happens in expected number of iterations = r). ■

Corollary 3.3.1. *If planted search* kSUM *is hard at density* Δ*, then there is no algorithm that runs in expected time* $O\left(r^{k/2\,(1-\Omega(1))}\,k^{-m}\right)$ *and solves the* targeted-vector-kSUM *problem at density* Δ *with advantage* $o(1)$.

Note that the Corollary 3.3.1 directly follows from Corollary 3.1.1 and the above reduction.

3.3 ALGORITHM FOR K-XOR AT LOW DENSITIES

In this section, unlike the above reductions, we will show that the k-XOR problem becomes easy to solve at densities $\Delta \leq \frac{1}{r^{0.5+\epsilon}}$. To start with, observe that the k-XOR problem becomes very easy to solve when the input matrix is square (i.e., $m = r$). This is because a random boolean square matrix is full rank with constant probability, in which case the planted k-XOR solution is the only linear dependence, and we can find it using Gaussian elimination in $O(r^3)$ time. This algorithm also works when $m > r$; we can simply ignore the bottom $m - r$ rows. Below, we will provide an algorithm for k-XOR that works by repeatedly trying to reduce a more general instance to this case.

Theorem 3.4 (Algorithm for k-XOR)**.** *For any* $k \in \mathbb{N}$*, there is an algorithm that runs in time* $\widetilde{O}\left(r^k m^{3-k}\right)$ *and solves the planted search* k*-XOR problem at any density with success probability* $1 - o(1)$.

Proof. Consider the following algorithm that is given input $\mathbf{A} \in \mathbb{F}_2^{m\times r}$. It becomes trivial when $r < m/2$, but in that case, the Gaussian elimination part can be applied directly to the instance to get the same results.

Algorithm $\mathcal{P}(\mathbf{A})$

1. Repeat the following steps $\left(\dfrac{4r}{m}\right)^k \log r$ times:

   1.1. Randomly choose $\frac{m}{2}$ columns of $\mathbf{A}$ to create a new $m \times \frac{m}{2}$ matrix $\mathbf{B}$.

   1.2. Run Gaussian elimination on $\mathbf{B}$ to find any linear relationships between its columns.

   1.3. If the previous step returns a linear dependence of size exactly k, return it.

2. Return $\perp$.

Observe that the planted solution is preserved in $\mathbf{B}$ in any given repetition with probability

$$\frac{\binom{r-k}{m/2-k}}{\binom{r}{m/2}} = \frac{(r-k)!\,(m/2)!}{(m/2-k)!\,r!} = \frac{(m/2)(m/2-1)\cdots(m/2-k+1)}{r\,(r-1)\cdots(r-k+1)} > \left(\frac{m/2-k+1}{r-k+1}\right)^k > \left(\frac{m}{4r}\right)^k$$

Since we run $(\frac{4r}{m})^k \log r$ repetitions, the probability that $\mathbf{B}$ contains the original k-XOR solution at least once is greater than

$$1 - \left(1 - \left(\frac{m}{4r}\right)^k\right)^{(\frac{4r}{m})^k \log r} > 1 - \frac{1}{r}$$

If we condition on $\mathbf{B}$ having the k-XOR solution preserved, it is easy to see that the induced distribution on $\mathbf{B}$ is the planted distribution on $\mathbb{F}_2^{m \times \frac{m}{2}}$. Since the number of rows is much larger than the number of columns, the planted solution is the only linear dependence in such a matrix with probability $1 - \mathrm{negl}(m)$. Therefore, Gaussian elimination returns a k-XOR solution, and we return the correct answer 1 with a probability greater than $1 - o(1)$.

The runtime of Gaussian elimination is $O(m^3)$. The total runtime of the above algorithm is therefore $O\left((\frac{r}{m})^k\, m^3 \log r\right) = \widetilde{O}(r^k\, m^{3-k})$. ∎

Corollary 3.4.1. *For any $k \in \mathbb{N}$ and constant $\epsilon > \dfrac{3}{k-3}$, there is an algorithm for the planted search k-XOR problem that runs in time $r^{\lceil k/2 \rceil - \Omega(1)}$.*

Proof. Since $m = \frac{k \log r}{\Delta}$, the runtime of $\mathcal{P}$

$$
\begin{aligned}
&= \widetilde{O}\left(\frac{r^k\, \Delta^{k-3}}{k^{k-3}}\right) \\
&= \widetilde{O}_k(r^{(k-3)/2}) \\
&= o(r^{\lceil k/2 \rceil - \Omega(1)})
\end{aligned}
$$

■

CHAPTER 4

K-SUM IN THE SPARSE REGIME

In this chapter, we show how assuming the hardness of kSUM (and its variant kXOR) in the sparse regime (i.e., when the density, $\Delta < 1$) helps us build various cryptographic primitives.

4.1 DEFINITIONS

Mixed-product property: Let $\mathbf{A}, \mathbf{B}, \mathbf{C}, \mathbf{D}$ be matrices that can form the matrix products $\mathbf{A}\,\mathbf{C}$ and $\mathbf{B}\,\mathbf{D}$ then,

$$(\mathbf{A} \otimes \mathbf{B}) \cdot (\mathbf{C} \otimes \mathbf{D}) = (\mathbf{A}\,\mathbf{C} \otimes \mathbf{B}\,\mathbf{D})$$

For any two vectors $\mathbf{u} \in \mathbb{F}_2^m, \mathbf{v} \in \mathbb{F}_2^n$, $(\mathbf{u} \otimes \mathbf{v} = \mathbf{u} \cdot \mathbf{v}^T) \in \mathbb{F}_2^{m \times n}$.

Definition 4.1 (Public Key Encryption). A *Public Key Encryption (*PKE*)* scheme for a message space $\mathcal{M}$ consists of PPT algorithms $\mathsf{PKE} = (\mathsf{KeyGen}, \mathsf{Enc}, \mathsf{Dec})$ with the following syntax:

- $\mathsf{KeyGen}(1^r) \to (\mathsf{pk}, \mathsf{sk})$: on input the unary representation of the security parameter r, generates a public key pk and a secret key sk.
- $\mathsf{Enc}(\mathsf{pk}, m) \to \mathsf{ct}$: on input a public key pk and a message $m \in \mathcal{M}$, outputs a ciphertext ct.
- $\mathsf{Dec}(\mathsf{sk}, \mathsf{ct}) \to m$: on input a secret key sk and a ciphertext ct, outputs a message $m \in \mathcal{M}$.

The scheme should satisfy the following properties:

Correctness. A scheme PKE is *correct* if there exists a negligible function negl such that for every security parameter r and message $m \in \mathcal{M}$:

$$\Pr[\mathsf{Dec}(\mathsf{sk}, \mathsf{Enc}(\mathsf{pk}, m)) = m] \geq 1 - \mathrm{negl}(r)$$

where $(\mathsf{pk}, \mathsf{sk}) \leftarrow \mathsf{KeyGen}(1^r)$. The scheme is *weakly* correct if the probability of correct decryption is bounded by $1 - o(1)$ instead of $1 - \mathrm{negl}(r)$ above.

CPA Security. A PKE scheme is $\mathsf{IND\text{-}CPA}$ *secure* if for any PPT adversary $\mathcal{A}$ there exists a negligible function negl such that:

$$\left|\Pr[\mathsf{PKEGame}^0_{\mathcal{A}(r)} = 1] - \Pr[\mathsf{PKEGame}^1_{\mathcal{A}(r)} = 1]\right| \leq \mathrm{negl}(r)$$

where $\mathsf{PKEGame}^b_{\mathcal{A}(r)}$ is a game between an adversary $\mathcal{A}$ and a challenger C with a challenge bit b defined as follows:

- C samples $(\mathsf{pk}, \mathsf{sk}) \leftarrow \mathsf{KeyGen}(1^r)$, and sends pk to $\mathcal{A}$.
- $\mathcal{A}$ chooses $m_0, m_1 \in \mathcal{M}$ and sends them to C.
- C computes $\mathsf{ct} \leftarrow \mathsf{Enc}(\mathsf{pk}, m_b)$, and sends ct to $\mathcal{A}$.
- The adversary $\mathcal{A}$ outputs a bit b' which we define as the output of the game.

The scheme is said to be *weakly* $\mathsf{IND\text{-}CPA}$ secure if we replace $\mathrm{negl}(r)$ with $o(1)$ in the condition above.

Definition 4.2 (Secret-key Somewhat Homomorphic Encryption). A *secret-key Somewhat Homomorphic Encryption (*SHE*)* scheme for a message space $\mathcal{M}$ consists of PPT algorithms $\mathsf{SHE} = (\mathsf{KeyGen}, \mathsf{Enc}, \mathsf{Dec}, \mathsf{Add}, \mathsf{Mult})$ with the following syntax:

- $\mathsf{KeyGen}(1^r) \to \mathsf{sk}$: on input the unary representation of the security parameter r, generates a secret key sk.
- $\mathsf{Enc}(\mathsf{sk}, m) \to \mathsf{ct}$: on input a secret key sk and a message $m \in \mathcal{M}$, outputs a ciphertext ct.
- $\mathsf{Dec}(\mathsf{sk}, \mathsf{ct}) \to m$: on input a secret key sk and a ciphertext ct, outputs a message $m \in \mathcal{M}$.

- $\mathsf{Add}(\mathsf{ct}_1, \mathsf{ct}_2) \rightarrow \mathsf{ct}_{add}$: on input two valid ciphertexts ct_1 and ct_2, outputs homomorphically added ciphertext ct_{add}.
- $\mathsf{Mult}(\mathsf{ct}_1, \mathsf{ct}_2) \rightarrow \mathsf{ct}_{mult}$: on input two valid ciphertexts ct_1 and ct_2, outputs homomorphically multiplied ciphertext ct_{mult}.

The scheme should satisfy the following properties:

Correctness. A scheme SHE is *correct* if it satisfies two types of correctness:

1. **Decryption correctness**. there exists a negligible function negl such that for every security parameter $r \in \mathbb{N}$ and message $m \in \mathcal{M}$.

$$\Pr[\mathsf{Dec}(sk, \mathsf{Enc}(\mathsf{sk}, m)) = m] \geq 1 - \mathrm{negl}(r)$$

where $(\mathsf{pk}, \mathsf{sk}) \leftarrow \mathsf{KeyGen}(1^r)$.

2. **Evaluation correctness**. For any two ciphertexts ct_1 and ct_2 generated as outputs of Enc (or) Add (or) Mult and encrypting messages m_1 and m_2 respectively.

$$\Pr[\mathsf{Dec}(\mathsf{sk}, \mathsf{Add}(\mathsf{ct}_1, \mathsf{ct}_2)) = m_1 + m_2] \geq 1 - \mathrm{negl}(r)$$
$$\Pr[\mathsf{Dec}(\mathsf{sk}, \mathsf{Mult}(\mathsf{ct}_1, \mathsf{ct}_2)) = m_1 \cdot m_2] \geq 1 - \mathrm{negl}(r)$$

The scheme is *weakly* correct if the probability of correct decryption (or) evaluation is bounded by $1 - o(1)$ instead of $1 - \mathrm{negl}(r)$ above. Further, the number of times that we can compose Add and Mult functions while generating homomorphic ciphertext decides the number of evaluations that can be performed by a scheme.

Security. An SHE scheme is (t, T) *secure*[1] if for any adversary $\mathcal{A}$ running in time T there exists a negligible function negl such that:

$$\left|\Pr[\mathsf{SHEGame}^0_{\mathcal{A}(r)} = 1] - \Pr[\mathsf{SHEGame}^1_{\mathcal{A}(r)} = 1]\right| \leq \mathrm{negl}(r)$$

where $\mathsf{SHEGame}^b_{\mathcal{A}(r)}$ is a game between an adversary $\mathcal{A}$ running in time T that can query atmost t many ciphertexts and a challenger C with a challenge bit b defined as follows:

[1]Note that if $t = O(r^{c_1})$ and $T = O(r^{c_2})$ for any constants c_1, c_2 then this is equivalent to IND-CPA. Note that we will prove IND-CPA security for our scheme.

- C samples $\mathsf{sk} \leftarrow \mathsf{KeyGen}(1^r)$.
- $\mathcal{A}$ chooses the challenge tuple $m_0, m_1 \in \mathcal{M}$ and sends them to C.
- C computes $\mathsf{ct} \leftarrow \mathsf{Enc}(\mathsf{pk}, m_b)$, and sends ct to $\mathcal{A}$.
- $\mathcal{A}$ can query upto t many ciphertexts. As a part of the query, $\mathcal{A}$ chooses a message μ_i and sends it to the challenger C to get the corresponding ciphertext ct_i. It can also query before the challenge tuple is selected.
- The adversary $\mathcal{A}$ outputs a bit b' which we define as the output of the game.

The scheme is said to be *weakly* (t, T) secure if we replace $\mathrm{negl}(r)$ with $o(1)$ in the condition above.

4.2 PUBLIC KEY ENCRYPTION FROM K-SUM AND LWE

In this section, we give a construction of a PKE scheme based on the hardness of kSUM and LWE.

Construction. Our public key encryption scheme $\mathsf{PKE} = (\mathsf{KeyGen}, \mathsf{Enc}, \mathsf{Dec})$ for message space $\mathcal{M} = \{0, 1\}$, is described as follows:

$\mathsf{PKE.KeyGen}(1^r)$: Upon input the unary representation of the security parameter r, do the following:

1. $\mathbf{A} \xleftarrow{\$} \mathbb{Z}_q^{m \times n}$
2. Sample $\mathbf{x} \in \{0, 1\}^n$ uniformly such that $\|\mathbf{x}\|_0 = k$
3. Let $\mathbf{u} := \mathbf{A}\,\mathbf{x}$
4. Output $\mathsf{pk} := (\mathbf{u}, \mathbf{A})$, $\mathsf{sk} := \mathbf{x}$

$\mathsf{PKE.Enc}(\mathsf{pk} := (\mathbf{u}, \mathbf{A}), \mu \in \{0, 1\})$: Upon input the public key pk and the message μ, do the following:

1. Compute $\mathbf{c}_1 := \mathbf{A}^T \mathbf{s} + \mathbf{e}$ where $\mathbf{s} \leftarrow \mathbb{Z}_q^m, \mathbf{e} \leftarrow \chi^n$

2. Compute $c_2 := \mathbf{u}^T \mathbf{s} + e' + \lfloor \frac{q}{2} \rfloor \mu$ where $e' \leftarrow \chi$

3. Output $\mathsf{ct} := (\mathbf{c}_1, c_2)$

$\mathsf{PKE.Dec}(\mathsf{pk} := (\mathbf{u}, \mathbf{A}), \mathsf{sk} := \mathbf{x}, \mathsf{ct} := (\mathbf{c}_1, c_2))$: Upon input the public key pk, the secret key sk and the ciphertext ct, do the following:

1. Compute $\boldsymbol{\mu}' := c_2 - \mathbf{x}^T \mathbf{c}_1$.

2. If $\|\boldsymbol{\mu}'\|_2 \leq \frac{q}{4}$ then output 0 else output 1.

Lemma 4.1 (Correctness)**.** *For security parameter $r \in \mathbb{N}$, given parameters of the* PKE *construction are set as mentioned in (Section 4.2). Then, the public key encryption scheme* PKE *described above is correct.*

Proof. Follows from a straightforward calculation.

$$\begin{aligned}
\boldsymbol{\mu}' &= c_2 - \mathbf{x}^T \mathbf{c}_1 \\
&= \mathbf{u}^T \mathbf{s} + e' + \left\lfloor \frac{q}{2} \right\rfloor \mu - \mathbf{x}^T \mathbf{A}^T \mathbf{s} - \mathbf{x}^T \mathbf{e}. \\
&= e' + \left\lfloor \frac{q}{2} \right\rfloor \mu - \mathbf{x}^T \mathbf{e}. \\
&= \left\lfloor \frac{q}{2} \right\rfloor \mu + (e' - \mathbf{x}^T \mathbf{e}).
\end{aligned}$$

Hence, when $\mu = 0$, the value is $e' - \mathbf{x}^T \mathbf{e}$, if we show that $\left\|e' - \mathbf{x}^T \mathbf{e}\right\|_2 \leq \frac{q}{4}$ then that would be sufficient.

Note that $\left\|\mathbf{x}^T \mathbf{e}\right\|_2 = k(\alpha q)$ with probability $1 - o(1)$, since at most k of the entries in the summation are from bounded distribution and the rest of them are zero, we have that $|e'| \leq \alpha \cdot q$ with probability $1 - o(1)$(Theorem 2.3), and so,

$$\left\|e' + \mathbf{x}^T \mathbf{e}\right\|_2 \leq (k+1)\alpha q$$

Therefore, the correctness holds for the parameters mentioned in Section 4.2. ∎

Next, we prove the security of our scheme under the hardness of LWE and k-SUM.

Lemma 4.2 (Security). *For security parameter $r \in \mathbb{N}$, assuming the hardness of planted-search-*kSUM *at density $\Delta = \frac{1}{\mathsf{polylog}(r)}$, and that* $\mathsf{LWE}(m, n+1, q, \chi)$*(Theorem 2.4) is $2^{\frac{m}{\log \frac{m}{\alpha}}}$-hard where the parameters are chosen as described in Section 4.2, the public key encryption scheme* PKE *satisfies weak* IND-CPA *security (Definition 4.1).*

Proof. We prove the theorem via a sequence of hybrids between the challenger and a PPT adversary $\mathcal{A}$.

Hybrid 0: This is the real world with challenge bit 0. This is same as $\mathsf{PKEGame}^0_{\mathcal{A}(r)}$

Hybrid 1: This world is same as Hybrid 0 except we sample public key pk as $(\mathbf{u}', \mathbf{A}) \xleftarrow{\$} (\mathbb{Z}_q^{n\times 1}, \mathbb{Z}_q^{m\times n})$.

Hybrid 2: This world is same as Hybrid 1 except we sample the ciphertext, i.e, ct := $(\mathbf{c}_1', c_2')$ as $\mathbf{c}_1' \xleftarrow{\$} \mathbb{Z}_q^n, c_2' \xleftarrow{\$} \mathbb{Z}_q$ respectively.

In Hybrid 2, the distribution seen by the adversary is independent of the challenge bit b; hence, the advantage of the adversary in this world is negligible.

Indistinguishability of Hybrids. We now show that consecutive hybrids are indistinguishable.

Claim 4.2.1. Assume that planted search kSUM is hard at density Δ (Corollary 3.3.1) for the parameters described in Section 4.2. Then, Hybrid 0 and Hybrid 1 are indistinguishable for any PPT adversary.

Proof. Let $\mathcal{D}$ be a PPT adversary that distinguishes Hybrid 0 from Hybrid 1 with advantage ϵ. Let C be the targeted-vector-kSUM challenger at density Δ. Then we give a reduction $\mathcal{B}$ that solves targeted-vector-kSUM with advantage ϵ as follows:

1. C outputs $(\mathbf{A}, \mathbf{u}^*) \in (\mathbb{Z}_q^{m\times n}, \mathbb{Z}_q^m)$.

2. $\mathcal{B}$ samples $\mu_0, \mu_1 \leftarrow \mathcal{M}, \mathbf{s} \xleftarrow{\$} \mathbb{Z}_q^m, \mathbf{e} \leftarrow \chi^n, e' \leftarrow \chi$

3. $\mathcal{B}$ then computes $(\mu_0, \mu_1, \mathbf{A}, \mathbf{u}^*, \mathbf{c}_1 := \mathbf{A}^T\mathbf{s} + \mathbf{e}, c_2 := \mathbf{u}^{*T}\mathbf{s} + e' + \lfloor \frac{q}{2} \rfloor \mu_0)$

4. If $\mathcal{D}$ outputs b then $\mathcal{B}$ outputs b

Notice that when C outputs $\mathbf{u}' \xleftarrow{\$} \mathbb{Z}_q^m$ then $\mathcal{B}$ simulates Hybrid 1 else, it simulates Hybrid 0. Hence, $\mathcal{B}$ solves targeted-vector-kSUM at density $\Delta = \frac{1}{\mathsf{polylog}(r)}$ with probability equal to the advantage of $\mathcal{D}$ i.e, ϵ.

From Corollary 3.3.1, $\epsilon = o(1)$. Therefore, by the hardness of targeted-vector-kSUM at density $\Delta = \frac{1}{\mathsf{polylog}(r)}$, we have Hybrid 0 $\approx_c$ Hybrid 1 ■

Claim 4.2.2. Assume that $\mathsf{LWE}(m, n+1, q, \chi)$ is $2^{\frac{m}{\log \frac{m}{\alpha}}}$ hard (Theorem 2.4) for the parameters described in Section 4.2. Then, Hybrid 1 and Hybrid 2 are indistinguishable for any PPT adversary.

Proof. Let $\mathcal{D}$ be a PPT adversary that distinguishes Hybrid 1 from Hybrid 2 with advantage ϵ. Let C be the $\mathsf{LWE}(m, n+1, q, \chi)$ challenger. Then we give a reduction $\mathcal{B}$ that solves $\mathsf{LWE}(m, n+1, q, \chi)$ problem with advantage ϵ as follows:

1. C outputs $\begin{bmatrix} \mathbf{c}_1^* \\ c_2^* \end{bmatrix} \in \mathbb{Z}_q^{n+1}$ such that $\mathbf{c}_1^* \in \mathbb{Z}_q^n$ and $c_2^* \in \mathbb{Z}_q$

2. $\mathcal{B}$ samples $\mu_0, \mu_1 \leftarrow \mathcal{M}, \mathbf{u}' \xleftarrow{\$} \mathbb{Z}_q^m, \mathbf{e} \leftarrow \chi^n, e' \leftarrow \chi$

3. $\mathcal{B}$ then computes $(\mu_0, \mu_1, \mathbf{A}, \mathbf{u}', \mathbf{c}_1^*, c_2^*)$

4. If $\mathcal{D}$ outputs b then $\mathcal{B}$ outputs b

Notice that when C outputs such that $\mathbf{c}_1^* \xleftarrow{\$} \mathbb{Z}_q^n$ and $c_2^* \xleftarrow{\$} \mathbb{Z}_q$ then $\mathcal{B}$ simulates Hybrid 2 else, it simulates Hybrid 1. Hence, $\mathcal{B}$ solves $\mathsf{LWE}(m, n+1, q, \chi)$ with probability equal to the advantage of $\mathcal{D}$ i.e, ϵ.

By the hardness of $\mathsf{LWE}(m, n+1, q, \chi)$ (Theorem 2.4) where χ is parameterised by αq, $\epsilon = \text{negl}(r)$. ■

From the claims (Claim 4.2.1 and Claim 4.2.2) we have that for any PPT adversary $\mathcal{A}$:

$$\left|\Pr[\mathsf{PKEGame}^0_{\mathcal{A}(r)} = 1] - \Pr[\mathsf{PKEGame}^1_{\mathcal{A}(r)} = 1]\right| \leq o(1) \qquad \blacksquare$$

Parameters. We now wish to give instantiations of parameters for the PKE construction in Section 4.2, and observe the improvements in these parameters due to replacing targeted-vector-kSUM as a computational analogue of "LHL" (Leftover Hash Lemma). Replacing LHL with targeted-vector-kSUM is equivalent to assuming mild planted-search-k-SUM (Corollary 3.3.1).

First, we enumerate the constraints that need to be satisfied for the correctness and security of PKE construction in Section 4.2. We then provide two instantiations of parameters along with the analysis of how assuming hardness of targeted-vector-kSUM assumption helps us improve the parameters of the underlying PKE scheme. For correctness to hold, from Theorem 4.1 we need $(k+1)\alpha q \leq \frac{q}{4}$ which implies that,

$$\alpha \leq \frac{1}{4(k+1)}.$$

For security wrt .Claim 4.2.2, we need the minimum time taken by an adversary to solve $\mathsf{LWE}(m, n, q, \chi)$ to be super-polynomial in r; this implies that,

$$2^{\frac{m}{\log \frac{m}{\alpha}}} \geq r^{\omega(1)}.$$

In order to use Theorem 2.4, we also need to satisfy the condition $q \geq 2\sqrt{m}k$. For security wrt Claim 4.2.1, we need that minimum time taken by an adversary to solve targeted-vector-kSUM at density Δ is super-polynomial in r. Thus, from Lemma 3.3.1, we require that $n^{\frac{k-2}{2}} k^{-m} \geq r^{\omega(1)}$.

Parameter Analysis. In the construction of PKE in Section 4.2, the $|\mathsf{pk}| = n\,m\ \log q$, $|\mathsf{sk}| = m\ \log q$, encryption time is $m\,k\ \log q$, decryption time is $m\ \log q$. Below, we provide parameters for two settings, one which emphasizes the improvement in the public key size and one which emphasizes the improvement of error rate (α) of $\mathsf{LWE}(m, n, q, \chi)$. Note that the first three constraints are to be satisfied irrespective of whether we assume the hardness of targeted-vector-kSUM (or) not. Therefore, we first set the parameters such that they satisfy the first three constraints, then focus only on the constraint that assuming targeted-vector-kSUM (or) using LHL enforces to analyze the improvement. Note that the constraint enforced by LHL is $m \leq \frac{k\ \log n}{\log q}$ (Theorem 2.2).

Parameters for Reducing Public Key Size. We basically fix all the values except n and see how it is affected by assuming targeted-vector-kSUM hardness in place of LHL. As the discussion is about efficiency in terms of pk-size, sk-size, encryption, and decryption time, when given a choice, we picked smaller values of m and q.

- $c = \omega(1)$
- $k = c\,(\log\log r)$
- $m = 2c\ \log r\,(\log\log r)$
- $q = 8\sqrt{c\,(\log r)\,(\log\log r)}\ (k+1)$
- $\frac{1}{\alpha} = 4\,(k+1)$

If we invoke LHL, i.e., try to satisfy the constraint $m \leq \frac{k\ \log n}{\log q}$ to argue security, then $n \geq r^2$. On the other hand if we assume hardness of targeted-vector-kSUM at density Δ then we have to satisfy the constraint that $\Delta = \frac{k \log n}{m\ \log q}$ instead and, assuming $\Delta = \frac{1}{\mathsf{polylog}(r)}$ means $n \geq (r)^{1/\log\log r}$. This means that the construction using targeted-vector-kSUM instead of LHL has considerable gain improvement in terms of pk-size, i.e.,

Targeted vector k-SUM construction	LHL-based construction
$\lvert\mathsf{pk}\rvert = r^{1/\log\log r}\,(2c\ \log r\ \log\log r)$	$\lvert\mathsf{pk}\rvert = r^2\,(2c\ \log r\ \log\log r)$

Table 4.1: Comparison of public key-size of vector-kSUM based construction and LHL based construction

Similarly, we now try to reduce the k value.

Parameters to Improve the error rate (α) of $\mathsf{LWE}(m, n, q, \chi)$. Setting $r = n$, would ease our analysis. As all the other parameters except k are the same, this assumption would not affect our qualitative analysis of how k is affected by the computational LHL. Therefore, we retain the values of q and m from the above analysis, and the parameters are as follows:

- $c = \omega(1)$
- $m = 2c\ \log r\,(\log\log r)$
- $q = 8\sqrt{c\ \log r\,(\log\log r)}\ (k+1)$
- $\frac{1}{\alpha} = 4\,(k+1)$
- $n = r$

Note that in LHL, one has to satisfy the condition of $m \leq \frac{k\log n}{\log q} \implies k \geq 2\,c(\log\log r)\ \log q$ whereas when we instantiate our scheme using targeted-vector-kSUM assumption, the condition to satisfy is $\Delta = \frac{k\ \log r}{m\log q}$. Notice that, in this case, we can set k as any super-constant because $\Delta = \frac{1}{\mathsf{polylog}(r)}$ which is fine. Thereby providing us with a small improvement in terms of the value of k. Note that by trading off n and k value, one can improve both error, $\lvert\mathsf{pk}\rvert$, and encryption time.

Assuming best known LWE algorithm runs in time 2^{m^c}: LWE is believed to be very robust. Unlike LPN where we know of an algorithm that runs in time $2^{\eta\cdot m}$ where η is the Bernoulli parameter of the noise, we don't know of any such algorithms for LWE. It is widely believed that the best known algorithm for LWE takes at least $2^{\frac{m}{\log\frac{m}{\alpha}}}$ time to solve.

However, one cannot completely ignore the possibility of finding better algorithms for LWE. In this spirit, we show that when we assume LWE to be 2^{m^c} hard for any $c < 0.5$ then we need k-SUM i.e,

Lemma 4.3 (Security). *For security parameter $r \in \mathbb{N}$, assuming the hardness of planted-search-*kSUM *at density* $\Delta = \frac{1}{\mathsf{polylog}(r)}$, *and that* $\mathsf{LWE}(m, n+1, q, \chi)$ *is* 2^{m^c}*-hard such that* $c < 0.5$ *where the parameters are chosen as described below, the public key encryption scheme* PKE *satisfies weak* IND-CPA *security (Definition 4.1).*

Proof. We begin by noting that this proof is identical to Theorem 4.2. The only change is in the assumption about running time of the best known LWE algorithm being 2^{m^c} hard for $c < 0.5$. Considering this change, the new constraints that need to be satisfied for the PKE (Section 4.2) to be correct and secure are as follows:

- $\alpha \leq \frac{1}{4(k+1)}$
- $2^{m^c} \geq r^{\omega(1)}$
- $q \geq 2\sqrt{m}k$
- $n^{\frac{k-2}{2}} k^{-m} \geq r^{\omega(1)}$

Note that after fixing any value such that $q \geq 2\sqrt{m}$, the constraint enforced by LHL $\left(m \leq \frac{k \log n}{\log q} \right)$ and the constraint enforced by LWE hardness $\left(2^{m^c} \geq r^{\omega(1)} \right)$ cannot be simultaneously satisfied. Therefore, we need k-SUM instead of LHL to prove security. The exact parameters required in this case while invoking k-SUM hardness are given below:

- $c = \omega(1)$
- $k = c\,(\log\log r)$
- $m = 2c(\log r\,(\log\log r))^{\frac{1}{c}}$
- $q = 2\sqrt{m}k$

- $\frac{1}{\alpha} = 4\,(k+1)$
- $n = r$

■

4.3 SOMEWHAT HOMOMORPHIC ENCRYPTION

In this section, we first define and give a flawed, somewhat homomorphic encryption scheme based on the hardness of the kXOR problem alone. This flawed scheme satisfies the correctness requirement of an SHE however, we point out a trivial attack on this scheme. To tackle such attacks, we conjecture a new hardness assumption inspired by the ideas that we saw in the study of the kSUM problem and argue why these new modifications avoid the trivial attacks mentioned earlier. We then proceed to give a construction of a somewhat homomorphic scheme based on the new hardness assumption.

4.3.1 Flawed Construction of Somewhat Homomorphic Encryption scheme

In this section, we adapt the construction in [BGV11] to come up with an SHE scheme (albeit flawed) based on hardness of the kXOR problem. We describe the (secret-key)[2] somewhat-homomorphic encryption scheme $\mathsf{SHE} = (\mathsf{Setup}, \mathsf{Enc}, \mathsf{Dec}, \mathsf{Add}, \mathsf{Mult})$ for message space $\mathcal{M} = \{0, 1\}$ below:

$\mathsf{SHE.Setup}(1^r)$: Upon input the unary representation of the security parameter r, do the following:

1. Set the parameters $n = r$, $k = \log\log r$, $m = (\log n)^c$ where $c > 1$ is any constant.
2. Sample $\mathbf{u} \leftarrow \{0,1\}^n_k$, $\mathbf{s} \xleftarrow{\$} \{0,1\}^m$ where k is an odd integer
3. Let $\mathbf{t} := \begin{bmatrix} -\mathbf{s} \\ 1 \end{bmatrix} \in \mathbb{F}_2^{(m+1)}$
4. Output $\mathsf{sk} := (\mathbf{t}, \mathbf{u})$

[2] When we say Somewhat Homomorphic Encryption throughout this work we mean secret-key Somewhat Homomorphic Encryption

$\mathsf{SHE.Enc}(\mathsf{sk} := (\mathbf{t}, \mathbf{u}), \mu \in \{0, 1\})$: Upon input the secret key sk and the message μ, do the following:

1. Sample $\mathbf{A} \xleftarrow{\$} \mathbb{F}_2^{n \times m}$, $\mathbf{e} \xleftarrow{\$} \{0, 1\}^n$
2. Replace $\mathbf{e}_j = \mu - \sum_{i \neq j : \mathbf{u}[i]=1} \mathbf{e}_i$, where $j = \min_{i:\mathbf{u}[i]=1} i$.
3. Compute $\mathbf{b} = \mathbf{A}\mathbf{s} + \mathbf{e}$[3]
4. Output $\mathsf{ct} := \left[\mathbf{A}\ \mathbf{b}\right] \in {\mathbb{F}_2}^{n \times (m+1)}$

$\mathsf{SHE.Dec}(\mathsf{sk} := (\mathbf{t}, \mathbf{u}), \mathsf{ct} := [\mathbf{A}\ \mathbf{b}])$: Upon input the secret key sk and the ciphertext ct, do the following:

1. Compute $\mathbf{c}' := \mathsf{ct} \cdot \mathbf{t}$
2. Compute $\mu' := \langle \mathbf{c}', \mathbf{u} \rangle$
3. Output μ'

The evaluation algorithm Add is given below:

$\mathsf{SHE.Add}(\mathsf{ct}_1, \mathsf{ct}_2)$: Upon input two ciphertexts ct_1 and ct_2, do the following:

1. Compute $\mathsf{ct}_{\mathsf{add}} := \mathsf{ct}_1 + \mathsf{ct}_2$
2. Output $\mathsf{ct}_{\mathsf{add}}$

$\mathsf{SHE.Mult}(\mathsf{ct}_1, \mathsf{ct}_2)$: Upon input two ciphertexts ct_1 and ct_2, do the following:

1. Compute $\mathsf{ct}_{\mathsf{mult}} := \mathsf{ct}_1 \otimes \mathsf{ct}_2$

Lemma 4.4 (Decryption Correctness)**.** *For security parameter $r \in \mathbb{N}$, given parameters of the* SHE *construction are set as mentioned in the setup. Then, the secret-key somewhat homomorphic encryption encryption scheme* SHE *has the property of decryption correctness.*

[3] s can be easily acquired from t

Proof. The decryption correctness follows from a straightforward calculation as follows:

$$\begin{aligned}\langle \mathbf{c}', \mathbf{u} \rangle &= \mathbf{e}^T \mathbf{u} \\ &= \mu\end{aligned}$$

■

Lemma 4.5 (Evaluation Correctness)**.** *For security parameter $r \in \mathbb{N}$, given parameters of the* SHE *construction are set as mentioned in the setup. Then, the secret-key somewhat homomorphic encryption encryption scheme* SHE *described above has the property of evaluation correctness.*

Proof. Given $\mathsf{ct}_1 := [\mathbf{A}_1\ \mathbf{b}_1]$ and $\mathsf{ct}_2 := [\mathbf{A}_2\ \mathbf{b}_2]$,

The evaluation correctness of the evaluated ciphertext(s) is as follows:

First we compute $\mathbf{t} := \begin{bmatrix} -\mathbf{s} \\ 1 \end{bmatrix}$ then we compute $\mathbf{c}' := \mathsf{Dec}(\mathsf{ct}_{add},\ \mathbf{t}) = -(\mathbf{A}_1 + \mathbf{A}_2)\mathbf{s} + (\mathbf{b}_1 + \mathbf{b}_2) = \mathbf{e}_1 + \mathbf{e}_2$ then,

$$\begin{aligned}\langle \mathbf{c}', \mathbf{u} \rangle &= \mathbf{c}'^T \mathbf{u} \\ &= \mathbf{e}_1^T \mathbf{u} + \mathbf{e}_2^T \mathbf{u} \\ &= \mu_1 + \mu_2\end{aligned}$$

Similarly for multiplication, we define $\mathbf{T} := (\mathbf{t} \otimes \mathbf{t})$ and $\mathbf{U} := (\mathbf{u} \otimes \mathbf{u})$, then we compute $\mathbf{C}' := \mathsf{Dec}(\mathsf{ct}_{mult}, (\mathbf{T}, \mathbf{U}))$.

Notice that,

$$\begin{aligned}\mathbf{C}' &= \mathsf{ct}_{mult}.\mathbf{T} \\ &= (\mathsf{ct}_1 \otimes \mathsf{ct}_2) \cdot (\mathbf{t} \otimes \mathbf{t}) \\ &= (\mathsf{ct}_1 \cdot \mathbf{t}) \otimes (\mathsf{ct}_2 \cdot \mathbf{t}) \text{ (By mixed-product property)}\end{aligned}$$

$$= (\mathbf{e}_1) \otimes (\mathbf{e}_2)$$

After that,

$$\begin{aligned}
\langle \mathbf{C}', \mathbf{U} \rangle &= \langle (\mathbf{e}_1 \otimes \mathbf{e}_2), (\mathbf{u} \otimes \mathbf{u}) \rangle \\
&= \langle (\mathbf{e}_1 \cdot \mathbf{e}_2^T), (\mathbf{u} \cdot \mathbf{u}^T) \rangle \\
&= (\mathbf{e}_1 \cdot \mathbf{e}_2^T)^T (\mathbf{u} \cdot \mathbf{u}^T) \\
&= (\mathbf{e}_2 \cdot \mathbf{e}_1^T) \cdot (\mathbf{u} \cdot \mathbf{u}^T) \\
&= (\mathbf{e}_2 \cdot \mu_1 \cdot \mathbf{u}^T) \\
&= \mu_1 \cdot \mu_2
\end{aligned}$$

■

Attack: While the above construction seems to be giving the correct functionality, it fails the security test. Ideally, we would like the SHE scheme to be $(\mathsf{poly}, \mathsf{poly})$ secure (Definition 4.2) i.e., the security of the scheme must hold even when the adversary is allowed to query polynomially many ciphertexts and run for any polynomial amount of time (in the security parameter), this notion is equivalent to the widely used IND-CPA security. But, wrt to the flawed scheme mentioned above, a PPT adversary that queries $O(nm)$ ciphertexts can easily come to know of the encrypted message using simple Gaussian elimination. The attack is as follows:

Given $\mathbf{A}_1, \mathbf{A}_2, .., \mathbf{A}_{nm} \in \{0, 1\}^{n \times m}$ are uniformly sampled (public) matrices.
Let us take a system of equations (which represent nm many ciphertexts of the msg '0'. These ciphertexts can be queried from the adversary as part of the security game):

$$\mathbf{A}_1\, \mathbf{s} + \mathbf{e}_1$$

$$\mathbf{A}_2\, \mathbf{s} + \mathbf{e}_2$$

$$\vdots$$

$$\mathbf{A}_{nm}\,\mathbf{s} + \mathbf{e}_{nm}$$

where $\mathbf{s} \in \{0,1\}^m$ (is the LPN secret) and the planted locations in each error ($\mathbf{e}_i$) that sum to 0 are secret (as per the SHE scheme given above).

Notice that one can easily find $\forall i \in [nm],\ x_i \in \{0,1\}$ such that $\sum_{i=1}^{nm} x_i\,\mathbf{A}_i = \mathbf{0}_{n\times m}$ by Gaussian elimination. This is because the entire $\mathbf{x} := (x_1,\ldots,x_{nm})$ can be thought of as a linear dependence among the $\mathbf{A}_i$'s. Because we are taking nm many ciphertexts, such an $\mathbf{x}$ exists with non-negligible probability. This is captured by the expected number of such $\mathbf{x}$ i.e,

$$\mathbb{E}[X] = \sum_{i=1}^{2^{nm}} \Pr[X_i = 1]$$

where $X = \sum_{i=1}^{2^{nm}} X_i$ is a random variable that represents number of such $\mathbf{x}$ and X_i is a random variable which takes the value 1 if the i^{th} combination[4] sums to the $\mathbf{0}$ matrix otherwise it is 0.

$$\begin{aligned}\mathbb{E}[X] &= \sum_{i=1}^{2^{nm}} \Pr[X_i = 1]\\ &= \sum_{i=1}^{2^{nm}} \frac{1}{2^{nm}}\\ &= 1\end{aligned}$$

Therefore, we can conclude that with a non-negligible probability there will exist a solution $\mathbf{x} = (x_1,\ \ldots,x_{nm})$ such that $\sum_{i=1}^{nm} x_i\,\mathbf{A}_i = \mathbf{0}_{n\times m}$ and this can be easily found by simple Gaussian elimination. Once we get this, then adding ciphertexts according to x_i's will give us the following:

$$\sum_{i=1}^{nm} x_i\big(\mathbf{A}_i\,\mathbf{s} + \mathbf{e}_i\big) = \sum_{i=1}^{nm} x_i\mathbf{e}_i$$

[4] We assume lexicographical ordering of all possible values of $\mathbf{x} \in \{0,1\}^{nm}$. For example 1^{st} combination means $\mathbf{x} = \vec{0}$

Thus, we end up with a sum of some e_i terms only. Now, if all the error vectors had k-XOR solution planted at a particular location, their sum would also have a k-XOR solution in the same location. Next, we repeat the entire procedure so far at least 'n' many times, and this yields a set of vectors that have the same k-XOR solution. But, now, because we have n k-XOR instances with a solution at the same location, it means that we have one k-XOR instance at density $\frac{k \log r}{n}$. At this density, k-XOR is easy to solve (Theorem 3.4). Therefore breaking the security of the above given SHE scheme.

4.3.2 New hardness assumption

With the intention to thwart any linear attacks on our scheme, we (informally) add an extra error to the ciphertext. The idea is that if one needs, say, t_1 ciphertexts to perform Gaussian elimination to break the security of our scheme, we will introduce the error such that there will be at least one erroneous ciphertext among any t_1 ciphertexts which will destroy any attack Gaussian elimination based attack. We capture this by introducing a new hardness assumption below.

Definition 4.3 (The Probabilistic-Planted-Error-LPN Problem). Let r be the security parameter, $n = r, m = m(r), k = k(n)$, be integers and, let $\eta = \eta(r)$. We define the $\mathsf{PPE-LPN}(m, n, k, \eta, i)$ problem as the task of distinguishing between the following distributions:

- $\{(\mathbf{A}_j, \mathbf{A}_j\mathbf{s} + \mathbf{e}_j)\}_{j \in [i]}$, where $\mathbf{A}_j \xleftarrow{\$} \mathbb{F}_2^{n \times m}$, $\mathbf{s} \xleftarrow{\$} \mathbb{F}_2^m$, and $\mathbf{e}_j \leftarrow \mathrm{ppBer}(\eta, k, n, \mathbf{u})$ where $\mathbf{u} \xleftarrow{\$} \{0, 1\}_k^n$.

- $\{(\mathbf{A}_j, \mathbf{v}_j)\}_{j \in [i]}$, where $\mathbf{A}_j \xleftarrow{\$} \mathbb{F}_2^{n \times m}$ and $\mathbf{v}_j \xleftarrow{\$} \mathbb{F}_2^n$

We say that the $\mathsf{PPE-LPN}(m, n, k, \eta, i)$ problem is *T-hard* if no adversary $\mathcal{A}$ running in time $T(n)$ can distinguish between the above distributions with advantage greater than $1/T(n)$.

An element from the distribution plantedBer$(\eta, k, n, \mathbf{u})$ is sampled as follows:

- $\mathbf{e} \leftarrow \mathrm{Ber}^n_{\frac{1}{2}}$.
- Sample $e' \leftarrow \mathrm{Ber}_\eta$
- Replace $\mathbf{e}_z = e' - \sum_{\ell \neq z : \mathbf{u}[\ell]=1} \mathbf{e}_\ell$, where $z = \min(\ell : \mathbf{u}[\ell] = 1)$.
- Output $\mathbf{e}$

Notation: $\forall\, j \in [i]$, we call each $(\mathbf{A}_j, \mathbf{A}_j\ \mathbf{s}{+}\mathbf{e}_j)$ as j^{th} sample of $\mathsf{PPE-LPN}(m, n, k, \eta, i)$ problem. A sample is called error-less (or) clear if $\langle \mathbf{e}_j, \mathbf{u} \rangle = 0$.

Conjecture 4.6. *For any security parameter $r \in \mathbb{N}$, $n = r$, $i = \mathsf{poly}(n)$, $k = k(n)$, and $m > \omega(1) \cdot \log r$ such that $k = \omega(1)$ is an odd integer, $\eta = \Omega(\frac{1}{(nm)})$ any PPT algorithm solves $\mathsf{PPE-LPN}(m, n, k, \eta, i)$ with $\mathrm{negl}(r)$ advantage.*

Reasoning for Conjecture 4.6

We have already shown above that having $O(nm)$ samples of the $\mathsf{PPE-LPN}$ problem (all without error) would let the adversary use Gaussian elimination to break the assumption. We conjecture that avoiding this kind of linear attack based on the planted linear relation should be enough because once the adversary cannot take advantage of the planted linear relation, then the error used in $\mathsf{PPE-LPN}$ can be intuitively thought of as uniform.

Let t_1 be the number of samples in clear (i.e., without error) that are required for the linear attack (current belief $t_1 = O(nm)$, for the sake of convenience, we take $t_1 = nm$).

The hope is that when we take any t_1 samples, there will be at least one erroneous sample with a high probability among them, which would thwart the Gaussian elimination based attack.

Let $X := \sum_{i \in [t_1]} X_i$ be an indicator random variable where $X_i = 1$ if the i^{th} sample is erroneous else 0.

$$\Pr[X = 0] = (1 - \eta)^{t_1}$$

The above equation basically gives the probability of zero erroneous samples among t_1 samples. Therefore, we need,

$$(1 - \eta)^{t_1} = \text{negl}(r)$$

which is true when $\eta = \Omega(\frac{1}{nm})$. We would like to have negl(r) also because we can then think of an adversary picking up any combination of nm samples again and again in the hope of an attack and none of them working.

4.3.3 Construction of a Somewhat Homomorphic Encryption Scheme

In this section, we use the newly conjectured hardness assumption and give a construction of an SHE scheme. The idea is to encrypt the wrong message with some probability so that when the adversary takes any set of more than $O(mn)$ ciphertexts to attack us using Gaussian elimination, then it will have at least one erroneous ciphertext, which will not let it use Gaussian elimination.

Let η be the probability with which we encrypt the wrong message. We describe our secret-key SHE = (Setup, Enc, Dec, Add, Mult) for message space $\mathcal{M} = \{0, 1\}$ below:

SHE.Setup(1^r): Upon input the unary representation of the security parameter r, do the following:

1. Set the parameters of the scheme according to the values given in Section 4.3.3
2. Sample $\mathbf{u} \leftarrow \{0, 1\}^n_k$, $\mathbf{s} \xleftarrow{\$} \{0, 1\}^m$ where k is an odd integer
3. Output sk := $(\mathbf{s}, \mathbf{u})$

SHE.Enc(sk := $(\mathbf{s}, \mathbf{u})$, $\mu \in \{0, 1\}$, coin): Upon input the secret key sk, the message μ, and a biased coin such that $\Pr[\text{coin} = 1] = \eta \in [0, 1]$ do the following:

For all $i \in [d]$:

1. Sample $\mathbf{A}_i \xleftarrow{\$} \mathbb{F}_2^{n\times m}$, $\mathbf{e}_i \xleftarrow{\$} \{0,1\}^n$
2. Replace $\mathbf{e}_{ij} = -\sum_{\ell\neq j:\mathbf{u}[\ell]=1} \mathbf{e}_{i\ell}$, where $j = \min_{\ell:\mathbf{u}[\ell]=1} \ell$.
3. According to the coin toss, if $\mathsf{coin} = 1$ then replace e_{ij} with $1 \oplus e_{ij}$[5]
4. Compute $\mathbf{b}_i = \mathbf{A}_i\mathbf{s} + \mathbf{e}_i + \mu[1\dots1]$
5. Compute $\mathsf{ct}_i := \left[\mathbf{A}_i\ \mathbf{b}_i\right] \in {\mathbb{F}_2}^{n\times(m+1)}$

Output $\mathsf{ct} := \{\mathsf{ct}_i\}_{i\in[d]}$

Note: We call the process of generating each ct_i as the base encryption procedure. This base encryption procedure is repeated 'd' times to enhance correctness of the scheme.

$\mathsf{SHE.Dec}(\mathsf{sk} := (\mathbf{s},\mathbf{u}), \mathsf{ct} := \{\mathsf{ct}_i\}_{i\in[d]})$: Upon input the secret key sk and the ciphertext ct, do the following:

1. Let $\mathbf{t} := \begin{bmatrix}-\mathbf{s}\\ 1\end{bmatrix} \in \mathbb{F}_2^{(m+1)}$
2. Compute for all $i \in [d]$, $\mathbf{c}'_i := \mathsf{ct}_i \cdot \mathbf{t}$
3. Compute for all $i \in [d]$, $\mu'_i := \langle \mathbf{c}'_i, \mathbf{u}\rangle$
4. Compute $\mu' = \sum_{i\in d} \mu'_i$. If $\mu' \geq \frac{d}{2}$ then output 1 else 0.

$\mathsf{SHE.Add}(\mathsf{ct}_1 := \{\mathsf{ct}_{1i}\}_{i\in[d]}, \mathsf{ct}_2 := \{\mathsf{ct}_{2i}\}_{i\in[d]})$: Upon input of two ciphertexts ct_1 and ct_2, do the following:

1. Compute $\mathsf{ct}_{\mathsf{add}} := \{\mathsf{ct}_{1i} + \mathsf{ct}_{2i}\}_{i\in[d]}$

$\mathsf{SHE.Mult}(\mathsf{ct}_1 := \{\mathsf{ct}_{1i}\}_{i\in[d]}, \mathsf{ct}_2 := \{\mathsf{ct}_{2i}\}_{i\in[d]})$: Upon input of two ciphertexts ct_1 and ct_2, do the following:

[5]Without this step, an adversary who got mn many ciphertexts could use Gaussian elimination to find the underlying encrypted message

1. Compute $\mathsf{ct}_{\mathsf{mult}} := \{\mathsf{ct}_{1i} \otimes \mathsf{ct}_{2i}\}_{i \in [d]}$

Below, we first set parameters and then argue about the correctness and security of the above construction.

Parameters

Using the bounds required for correctness and security (see below), we set the parameters as follows:

- $n = r$ where r is the security parameter of the scheme.
- m can be any value such that $m \geq \omega(1) \cdot \log r$.
- $\eta = \frac{1}{(t_1)^{0.9}}$ where η denotes the probability with which wrong message is encrypted.
- $t_1 = O(mn) \approx n \log n \cdot \omega(1)$, $d = \omega(1) \cdot \log n$. where t_1 denotes the minimum number of error less ciphertexts required to mount the Gaussian elimination based attack.
- $t_2 < (t_1)^{0.9}$ where t_2 denotes total number of evaluations possible.

Correctness

First, we will look at the decryption correctness of a single ciphertext and then proceed to show correctness for ciphertext formed after 't_2' number of evaluations under evaluation correctness.

Lemma 4.7 (Decryption correctness). *For any $r \in \mathbb{N}$, given parameters of the secret-key* SHE *are set as mentioned in (Section 4.3.3), the secret key somewhat homomorphic scheme* SHE *described above satisfies the decryption correctness property (as mentioned in Definition 4.2).*

Proof. Let us look at $\mathsf{ct} := \{\mathsf{ct}_i\}_{i \in [d]}$. Note that among $\{\mathsf{ct}_i\}_{i \in [d]}$ if there are at least $\frac{d}{2}$ ciphertexts (i.e, ct_i) that correctly encrypt the message ($i.e, \mathsf{coin_i} = 0$) then the decryption correctness is trivial as shown in Theorem 4.4. Therefore, it suffices to show that the probability of the number of erroneous ciphertexts being greater than $\frac{d}{2}$ is $\mathrm{negl}(r)$.

Let $Y := \sum_{i\in[d]} Y_i$ be an indicator random variable where $Y_i = 1$ if ct_i is erroneous else 0.

Note: Y_i, Y_j for any $i \neq j$ are independent because the erroneous nature of the ciphertext depends on their respective coin tosses which are independent. Therefore, we can apply Chernoff bound (Theorem 2.1) on this, and we get,

$$\Pr[Y > \frac{d}{2}] < e^{-\frac{(1-2\eta)^2\cdot d}{8\eta}}$$

Note that this is a bound on the probability of incorrect encryptions (because it's a bound on more than half of the ciphertexts being erroneous).

Substituting the parameters as mentioned in Section 4.3.3 we get,

$$e^{-\frac{(1-2\eta)^2\cdot d}{8\eta}} = \mathrm{negl}(r)$$

■

Evaluation correctness

Lemma 4.8 (Evaluation correctness). *For any $r \in \mathbb{N}$, given parameters of the secret-key* SHE *are set as mentioned in (Section 4.3.3), the secret key somewhat homomorphic scheme* SHE *described above has the following properties:*

- *It can correctly decrypt any ciphertext that is generated homomorphically using at most $t_2 = (mn)^{0.9}$ many evaluations (both addition and multiplication combined).*
- *Out of the t_2 many evaluations, it can support at most $t_{mult} := z$ many multiplications and the size of the ciphertext generated is $(nm)^z$ where z is any constant.*

Proof. Let t_2 be the number of evaluations that we can support. For our understanding, let us first take $d = 1$, note that when we homomorphically evaluate t_2 ciphertexts, to get a correct output, the evaluated ciphertext ct_{eval} must have no error, this is possible if all t_2 ciphertexts have no errors. Once this condition is satisfied then the evaluation correctness is trivial as shown in Theorem 4.5.

Using this idea, let us take the case of $d > 1$, we can visualize it as shown in Fig. 4.1 below. Let each column indicate a ciphertext ct with its d copies, and each row represents the set of ciphertexts that are combined homomorphically to generate a copy of the evaluated ciphertext. Therefore, there are t_2 columns below as we are evaluating t_2 ciphertexts.

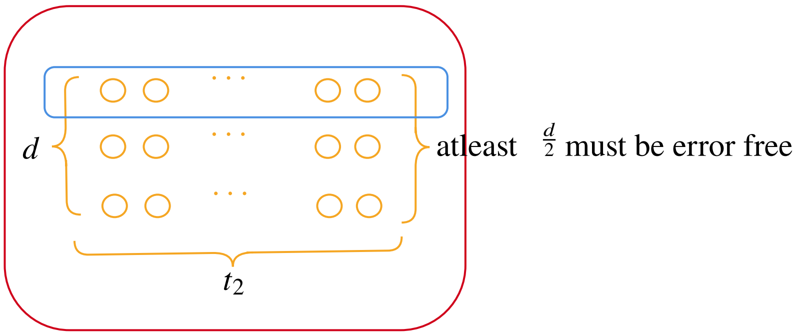

Figure 4.1: Correctness-requirement for evaluation

Let p be the probability of an individual evaluated ciphertext (i.e., evaluation of a row of ciphertexts) having an error, i.e.,

$$p = 1 - (1 - \eta)^{t_2}$$

Similar to the correctness of decryption, it suffices to show that the probability of the number of erroneous evaluated ciphertexts being greater than $\frac{d}{2}$ is $\mathsf{negl}(r)$.

Let $Y' := \sum_{i \in [d]} Y'_i$ be an indicator random variable where $Y'_i = 1$ if the corresponding evaluated ciphertext ct'_i is erroneous else 0.

Note: Y'_i, Y'_j for any $i \neq j$ are independent. Therefore, we can apply Chernoff bound

(Theorem 2.1) on this, and we get,

$$\Pr[Y' > \frac{d}{2}] < e^{-\frac{(1-2p)^2 \cdot d}{8p}}$$

Substituting the parameters as mentioned in Section 4.3.3 we get,

$$e^{-\frac{(1-2p)^2 \cdot d}{8p}} = \mathrm{negl}(r)$$

While we have shown above that a total of atmost $t_2 = (nm)^{0.9}$ many homomorphic evaluations can be performed. We note that multiplication operation results in quadratic blowup in the dimension size. i.e, $\mathsf{ct}_{mult} := \mathsf{ct}_1 \otimes \mathsf{ct}_2 \in \{0, 1\}^{(nm)^2}$. Therefore, if we were to perform t_{mult} many multiplications then the size of the ciphertext will be $(nm)^{t_{mult}}$ hence, t_{mult} must be a constant to make sure that the ciphertext is polynomial sized in the security parameter. ■

Security

Lemma 4.9 (Security). *Assuming the hardness of* PPE − LPN *(Conjecture 4.6), where the parameters are chosen as described in Section 4.3.3, the secret-key* SHE *satisfies* IND-CPA *security*

Proof. We prove the theorem via a sequence of hybrids between the challenger and a PPT adversary $\mathcal{A}$.

Hybrid 0: This is the real world with challenge bit b. This is same as $\mathsf{IND\text{-}CPA}^{b}_{\mathcal{A}(r)}$

Hybrid 1: This world is same as Hybrid 0 except we sample the ciphertexts (both challenge and query ciphertexts) uniformly, i.e, $\mathsf{ct} := \{[\mathbf{C}_{1j}, \mathbf{c}_{2j}]\}_{j \in [d]}$ as $\mathbf{C}_{1j} \xleftarrow{\$} \mathbb{Z}_q^{n \times m}$, $\mathbf{c}_{2j} \xleftarrow{\$} \mathbb{Z}_q^{n}$ for all $j \in [d]$.

In Hybrid 1, the distribution seen by the adversary is independent of the challenge bit b; hence, the advantage of the adversary in this world is negligible. ■

Indistinguishability of Hybrids. We now show that consecutive hybrids are indistinguishable.

Claim 4.9.1. Assume that $\mathsf{PPE-LPN}$ is hard (Conjecture 4.6) for the parameters described in Section 4.3.3. Then, Hybrid 0 and Hybrid 1 are indistinguishable for any PPT adversary.

Proof. Let $\mathcal{D}$ be a PPT adversary that distinguishes Hybrid 0 from Hybrid 1 with advantage ϵ. Let C be the PPE-LPN challenger. Let $i - 1 = \mathsf{poly}(r)$ be the total number of ciphertext queries asked in the IND-CPA game. Then we give a reduction $\mathcal{B}$ that solves PPE-LPN with advantage ϵ as follows:

Reduction $\mathcal{B}$:

1. C outputs $\{(\mathbf{A}_j, \mathbf{u}_j^*) \in (\mathbb{Z}_q^{n\times m}, \mathbb{Z}_q^n)\}_{j\in[i.d]}$ as a part of the $\mathsf{PPE-LPN}$ challenge to $\mathcal{B}$.
2. $\mathcal{B}$ samples $\mu_0, \mu_1 \leftarrow \mathcal{M}$
3. $\mathcal{B}$ then computes the challenge ciphertext ct^* as $\mathsf{ct}^* := \{\mathbf{A}_j, \mathbf{u}_j^* + \mu_b[1 \ldots 1]^T\}_{j\in[d]}$
4. $\mathcal{B}$ also computes the z^{th} query ciphertext ct'_z on the plaintext μ_z as $\mathsf{ct}'_z := \{\mathbf{A}_j, \mathbf{u}_j^* + \mu[1 \ldots 1]^T\}_{j\in[z\cdot d,\ (z+1)\cdot d]}$
5. $\mathcal{B}$ then sends $(\{(\mu_z, \mathsf{ct}'_z)\}_{z\in[i-1]}, (\mu_0, \mu_1, \mathsf{ct}^*))$ to the adversary $\mathcal{A}$.
6. If $\mathcal{A}$ outputs b' then $\mathcal{B}$ outputs b'

Notice that when C outputs $\{\mathbf{u}_j^* \xleftarrow{\$} \mathbb{Z}_q^m\}_{j\in[i\cdot d]}$ then $\mathcal{B}$ simulates Hybrid 1 else, it simulates Hybrid 0. Hence, $\mathcal{B}$ solves PPE-LPN with probability equal to the advantage of $\mathcal{D}$ i.e, ϵ.

From Conjecture 4.6, $\epsilon = \mathrm{negl}(r)$. Therefore, by the hardness of PPE-LPN, we have Hybrid 0 $\approx_c$ Hybrid 1

In Hybrid 1, the distribution seen by the adversary is independent of the challenge bit b;

Therefore, we have that for any PPT adversary $\mathcal{A}$:

$$\left|\Pr[\textsf{IND-CPA}^0_{\mathcal{A}(r)} = 1] - \Pr[\textsf{IND-CPA}^1_{\mathcal{A}(r)} = 1]\right| \leq \text{negl}(r) \qquad \blacksquare$$

Compactness

An important property for an SHE is compactness (wrt the size of the ciphertext). According to [Hal17], a homomorphic encryption scheme is uninteresting if it cannot beat the trivial homomorphic in terms of size of ciphertext. A trivial (homomorphic) scheme is one that gives out all the ciphertexts and function description and expects the client to decrypt all the ciphertexts and evaluate the function on the decrypted messages. In this section, we show that our scheme is better than the trivial scheme in terms of compactness.

Our SHE scheme also satisfies the non-trivial notion of compactness as described in [MGH08], which allows the ciphertext to grow exponentially with the multiplications but asks for the support of additions without increase in ciphertext size.

Consider the following trivial scheme: Use any standard Secret Key Encryption (SKE) scheme for Keygen and Encryption. To evaluate the ciphertexts $\textsf{ct}_1, \textsf{ct}_2, \ldots, \textsf{ct}_k$ homomorphically wrt the function f, simply concatenate all the ciphertexts and the description of the function circuit and essentially return the tuple $(f, \textsf{ct}_1, \textsf{ct}_2, \ldots, \textsf{ct}_k)$. Let decryption be an algorithm that, given a tuple (of function description and ciphertexts), decrypts all but the first element, parses the first element as a function, and then evaluates that function on the decrypted plaintexts as inputs.

In this trivial scheme, let us say we support at most s evaluations, this means that we can support s gates in the function that can be homomorphically computed. The corresponding circuit of the function, which has s gates, can be described using $O(s \log s)$ bits. This is because, one can think of the circuit as a DAG (in fact, it is undirected because of no feedback into a gate), and we can represent the DAG using all the edges. Note that we only need $2 \log s$ bits to represent an edge (each node is a gate and needs $\log s$ bits to represent). Note that the maximum number of edges in an acyclic graph is $(\#nodes - 1)$. Therefore, the total number of bits required to represent a circuit is at most $2s \log s$ bits.

Assuming each fresh ciphertext ct_i is of size $\mathsf{polylog}(n)$[6] (where $n = r$ is the security parameter), a trivial SHE scheme of giving out all ciphertexts and the function description requires size $2 \cdot \mathsf{polylog}(n) \cdot s + O(s \log s)$ (ciphertext + function description) where s is the number of ciphertexts that we can support.

Note: Our SHE scheme's ciphertext size is nm (If we consider only addition then our evaluated ciphertext supporting roughly nm many additions is also of the size nm), whereas the trivial ciphertext size supporting roughly nm many additions is $\Omega(nm \cdot \mathsf{polylog}(n))$. Hence, our scheme outperforms the trivial scheme in terms of compactness by a factor of $\mathsf{polylog}(n)$.

Limitations with our present construction

Compactness. We note that the blowing up of the dimension of ciphertext from $O(nm)$ to $O((nm)^2)$ on multiplication is a cause of concern as it restricts the possible number of multiplications to a constant.

The BV-FHE scheme [BGV11] similarly blows up the dimension of the ciphertext after one multiplication evaluation. But they manage to bring it back by an idea called dimension switching, which reduces the dimension of the homomorphically computed ciphertext but at the cost of introducing a new error, let us call e_{dr}. We sketch the idea

[6] The smallest known size of the ciphertext from LPN based assumptions [Ale03a] of a bit b can be realised as follows : $(\mathbf{a}, \langle \mathbf{a} \cdot \mathbf{s} \rangle + \mathbf{e} + b)$ such that $\mathbf{a} \xleftarrow{\$} \{0, 1\}^t$, $\mathbf{e}$ is the Bernoulli error vector (randomness used in encryption), $\mathbf{s} \xleftarrow{\$} \{0, 1\}^t$(secret key) where where $t = \mathsf{polylog}(n)$.

below and informally discuss why this idea cannot be used in its current form here.

Let $\mathsf{ct}_{\mathsf{mult}}$, $\mathsf{sk} = \mathbf{t} \otimes \mathbf{t}$ as defined above. The idea is to construct a matrix $\mathbf{B}$ (of order $(m+1)^2 \times (m+1)) \ni \mathbf{B} \cdot \mathbf{t} \approx \mathbf{t} \otimes \mathbf{t}$ then according to [BGV11], we take new ciphertext of smaller dimension $\mathsf{ct}' \triangleq \mathsf{ct}_{\mathsf{mult}} \cdot \mathbf{B}$, such that:

$$\begin{aligned}\mathsf{ct}' \cdot \mathbf{t} &\approx \mathsf{ct}_{\mathsf{mult}} \cdot \mathbf{B} \cdot \mathbf{t} \\ &\approx \mathsf{ct}_{\mathsf{mult}}(\mathbf{t} \otimes \mathbf{t}) \\ &\approx \mu_1 \otimes \mu_2 (\text{ By matrix product property})\end{aligned}$$

The idea in [BGV11] works because the ciphertext there is a vector, whereas we cannot afford the ciphertext to be a vector because of our planting. The fact that the ciphertext is not a vector means that even after multiplying with the hint matrix $\mathbf{B}$ on one side of the evaluated ciphertext $\mathsf{ct}_{\mathsf{mult}}$, the dimension of ct' still turns out to be n^2, which is not helpful and moreover, we cannot multiply anything to the left of the ciphertext because that would mean destroying the planted solution in error. Therefore, it seems unlikely that the idea of dimension reduction will yield any benefit, and hence, the need to look for new techniques to achieve compactness arises in our case.

It is worth noting that the nontrivial compactness as shown above should be seen as a step in the direction to build Homomorphic Encryption for bigger class of circuits using different assumptions like LPN similar to how [MGH08] was for the lattice problems based Homomorphic Encryption. Ideally, we would like the class of circuits supported to be more complex than the decryption circuit, which unfortunately our scheme does not satisfy. This is because the decryption for our scheme requires $(\omega(1)(\log n)nm)$ many additions and 1 comparison [7]. This is already more than the number of evaluations that

[7] For one $\mathsf{ct} = (\mathbf{A}, \mathbf{A}\,\mathbf{s} + \mathbf{e})$, we do $(\mathbf{A}\,\mathbf{s} + \mathbf{e}) - \mathbf{A}\mathbf{s}$ (This requires adding at most 'm' columns of $\mathbf{A}$ (where each column has 'n' bits) and then one subtraction (which is equivalent to one addition) and then we add 'k' elements of 'e' (which takes '$k = \omega(1) \log n$' additions). This base decryption needs to be done for 'd' $= \omega(1) \log n$ many ciphertexts as we are encrypting each message 'd' many times for the correctness and then take a majority of the values from 'd' many ciphertext decryptions

our scheme can support i.e, $(nm)^{0.9}$. Therefore, the above requirement about complexity of decryption circuit cannot be satisfied. Therefore, one of the important questions to ask for the future would be " *can our decryption circuit be simplified to be made of smaller size than the class of computations supported?* ".

4.4 UNIVERSALLY COMPOSABLE OBLIVIOUS TRANSFER FROM LPN AND K-XOR

Oblivious transfer (OT) [Wie83, Rab05, EGL85] is an important cryptographic primitive in which there exists a sender who has two input bits x_0 and x_1, and a receiver who chooses which bit x_b she wants to learn. On one hand, the receiver should learn x_b, but should have no information about $x_{1\oplus b}$. On the other hand, the sender should not learn the choice bit b. The Universally Composable Oblivious Transfer (UC-OT) primitive considers security against active static adversaries. That is, the adversaries can deviate in any arbitrary fashion from the protocol but can only corrupt the parties involved in the protocol (sender and receiver) at the beginning of the protocol. It is possible to realize the functionality that we need only in the Common Reference String (CRS) model [DDN14], where all the parties in the protocol are assumed to have access to a fixed string generated before protocol execution starts. In this work, we make use of the core idea used in [DDN14] i.e., using LPN-based IND-CPA secure public key cryptosystem as a building block we construct an extractable commitment scheme which implies a UC-OT protocol.

Theorem 4.10 ([DDN14])**.** *For a security parameter $r \in \mathbb{N}$, If there exists an extractable commitment scheme that is secure against PPT adversaries, then there exists an UC-OT protocol secure against active static PPT adversaries.*

Note: We only show that we can construct an extractable commitment scheme which directly implies the existence of UC-OT protocol due to the above theorem[8].

4.4.1 Encryption

In this section, we give the construction of a PKE scheme based on the hardness of kXOR and LPN, as shown in the paper ([ASS+23]). This construction is not part of this thesis but is written as an aid to understanding our Universally Composable Oblivious Transfer (UC-OT) protocol.

Construction. The public key encryption scheme $PKE_{\eta,k,\ell,m} = (\mathsf{KeyGen}, \mathsf{Enc}, \mathsf{Dec})$ for message space $\mathcal{M} = \{0, 1\}$, is described as follows:

$PKE_{\eta,k,\ell,m}.\mathsf{KeyGen}(1^r)$: Upon input the unary representation of the security parameter r, do the following:

1. Let $\mathsf{pk} \leftarrow \mathbb{F}_2^{m\times r}$.
2. Choose a random set $\mathsf{sk} \subseteq [r]$ with $|\mathsf{sk}| = k$.
3. Let $i \in \mathsf{sk}$ be the smallest index and let $\mathsf{pk}_i \leftarrow -\sum_{\substack{j\in\mathsf{sk}\\ j\neq i}} \mathsf{pk}_j$.
4. Return $\langle \mathsf{pk}, \mathsf{sk} \rangle$.

Here, we will interpret each $\mathsf{sk} \in \mathbb{F}_2^r$ as the characteristic vector for the set $\mathsf{sk} \subseteq [r]$.

$PKE_{\eta,k,\ell,m}.\mathsf{Enc}(\mathsf{pk}, b \in \{0, 1\})$: Upon input the public key pk and the message b, do the following:

2.1. Let $\mathbf{S} \leftarrow \mathbb{F}_2^{\ell\times m}$.

2.2. Let $\mathbf{E} \leftarrow \mathrm{Ber}_\eta^{\ell\times r}$.

2.3. Return $\mathbf{C} \leftarrow \mathbf{S}\,\mathsf{pk} + \mathbf{E} + \mathbf{B}$[9].

[8] As we only construct an extractable commitment scheme, we refrain from formally defining UC-OT, for this please refer to [DDN14].

[9] B:=a matrix where all elements are b, one can think of this as an encoding of message bit b

$PKE_{\eta,k,\ell,m}.\mathsf{Dec}(\mathsf{pk}, \mathsf{sk}, \mathsf{ct} := \mathbf{C})$: Upon input the public key pk, the secret key sk and the ciphertext ct, do the following:

1. Return 0 if $\|\mathbf{C}\,\mathsf{sk}\|_0 > \ell\left(\frac{1}{2} - \frac{(1-\eta^k)}{4}\right)$; else return 1.

Theorem 1.2 (PKE from k-XOR and LPN [ASS+23])**.** *There are values of $\eta = \Theta(1)$, $k = O(\log r)$, $m = r^{o(1)}$, and $\ell = \mathsf{poly}(r)$ such that the cryptosystem $PKE_{\eta,k,\ell,m}$ is a (weak) public-key bit encryption scheme, with all operations running in time $r^{1+o(1)}$, if any of the following conditions are satisfied:*

1. *Constant-noise LPN is 2^{m^c}-hard (for any constant $c > 0$) and planted search k-XOR is $r^{k/2-o(1)}$-hard[10] at every density $\frac{1}{\mathsf{polylog}(r)}$;*

4.4.2 Commitment Scheme

A commitment scheme allows Alice to "commit" to a message m by sending a commitment value com, such that the commitment has the following properties:

- **Hiding**: the commitment reveals nothing about m.
- **Binding**: Once committed, a commitment cannot be "opened" to a different message.

More formally,

Definition 4.4 (Commitment Scheme [KL07]). A *Commitment Scheme* scheme for a message space $\mathcal{M}$ consists of PPT algorithms $\mathsf{Gen}, \mathsf{Com}, \mathsf{Open}$ with the following syntax:

- $\mathsf{Gen}(1^r) \to params$: on input the unary representation of the security parameter r, generates parameters $params$ used by the scheme.
- $\mathsf{Com}(m \in \mathcal{M}) \to \mathsf{com}, r$: on input a message $m \in \mathcal{M}$, outputs a commitment com and randomness r used to commit.
- $\mathsf{Open}(\mathsf{com}, r) \to \{m, \perp\}$: on input a commitment com and randomness r, outputs a message $m \in \mathcal{M}$ (or) ivalid output $\perp$.

The scheme should satisfy the following properties:

[10] Informally, it means that no algorithm running in time $r^{k/2-o(1)}$ can solve the k-XOR problem at the given density

Hiding. $\Pr[\mathsf{Hiding}_{\mathcal{A},\mathsf{Com}}(1^\lambda) = 1] \leq \frac{1}{2} + \mathrm{negl}(n)$

Binding. $\Pr[\mathsf{Binding}_{\mathcal{A},\mathsf{Com}}(1^\lambda) = 1] \leq \frac{1}{2} + \mathrm{negl}(n)$

The commitment hiding game $\mathsf{Hiding}_{\mathcal{A},\mathsf{Com}}(1^\lambda)$ of the scheme Com against an adversary $\mathcal{A}$ is defined as follows:

- Parameters $params \leftarrow \mathsf{Gen}(1^\lambda)$ are generated.
- The adversary $\mathcal{A}$ is given input $params$, and outputs a pair of messages $m_0, m_1 \in \mathcal{M}$.
- $b \xleftarrow{\$} \{0,1\}$, r be the uniformly sampled randomness. Then compute the commitment $(\mathsf{com}, r) \leftarrow \mathsf{Com}(params, m_b)$.
- The adversary $\mathcal{A}$ is given com and outputs a bit b'.
- The output of the experiment is 1 if and only if $b' = b$.

The commitment binding game $\mathsf{Binding}_{\mathcal{A},\mathsf{Com}}(1^\lambda)$ of the scheme Com against an adversary $\mathcal{A}$ is defined as follows:

- Parameters $params \leftarrow \mathsf{Gen}(1^\lambda)$ are generated.
- The adversary $\mathcal{A}$ is given input $params$, and outputs $(\mathsf{com}, m, r, m', r')$.
- The output of the experiment is defined to be 1 if and only if $m \neq m'$ and $\mathsf{Com}(params, m; r) = \mathsf{com} = \mathsf{Com}(params, m'; r')$.

An Extractable Commitment Scheme

A commitment scheme is said to be extractable if there exists a polynomial-time simulator that is able to obtain the committed value m before the Open phase. In the CRS model, we will build an extractable commitment scheme based on the LPN and kXOR based public encryption scheme seen in Theorem 1.2. The commitment scheme is as follows:

We describe the commitment scheme COM for a message space $\mathcal{M}$ consisting of PPT algorithms $\mathsf{Gen}, \mathsf{Com}, \mathsf{Open}$ as follows:

$\mathsf{Gen}(1^r)$: Upon input the security parameter r, sample $\mathsf{pk} \xleftarrow{\$} \mathbb{F}_2^{m\times r}$ and output it.

$\mathsf{Com}_{\mathsf{pk}}(b \in \mathcal{M})$: Upon input the message m, do the following:

1. Encrypt m under the public key pk and let the randomness generated during the encryption be represented by $(\mathbf{S}, \mathbf{E})$ where $\mathbf{S}$ represents the LPN-secret and $\mathbf{E}$ represents the Bernoulli error matrix and outputs the corresponding ciphertext $\mathsf{ct} := \mathbf{S}\mathsf{pk} + \mathbf{E} + \mathbf{B}$ as a commitment com.

$\mathsf{Open}_{\mathsf{pk}}(\mathsf{com}, \mathbf{S}, \mathbf{E})$: Upon input the commitment com and randomness $(\mathbf{S}, \mathbf{E})$ used in the commitment do the following:

1. Check if $\mathsf{com} - \mathbf{E} \stackrel{?}{=} \mathbf{S}\,\mathsf{pk} + \mathbf{E}$

2. If it passes the above check, then check if for the majority of columns of $\mathbf{E}$ (i.e., $\geq \frac{\ell}{2}$), the hamming weight is bounded by $\eta \cdot r$.

3. If it passes both the tests then output $m := \mathsf{com} - \mathbf{E} - \mathbf{S}\,\mathsf{pk}$.

Claim 4.10.1. For the security parameter $r \in \mathbb{N}$, there are values of $\eta = \Theta(1)$, $k = O(\log r)$, $m = r^{o(1)}$, and $\ell = \mathsf{poly}(r)$ such that assuming constant-noise LPN is 2^{m^c} hard for $c < 0.5$ and that k-XOR is hard at density $\frac{1}{\mathsf{polylog}(r)}$, the construction mentioned above (i.e., COM) is an extractable commitment scheme over the message space $\mathcal{M}$.

Proof. Replicating the proof given in [DDN14] to our setting, we note that if both the parties (The committer and the receiver) are honest, then the hamming weight test will be passed with all but negligible probability. This is because the check on bound of hamming weights fails only when decryption correctness fails, and this is negligible due to the correctness of the underlying PKE scheme (Theorem 1.2). Similarly, the first test is trivially passed by honest parties.

The *hiding property* follows from the IND-CPA security of the encryption scheme. For the *binding property*, first notice that the hamming weight test performed during the opening phase ensures that the error term is within the bound such that the decryption is successful and the committed message m is decrypted. Note that due to the uniqueness of the decryption, there is only one m that can pass both the tests performed in the opening phase.

In order to extract the committed values and thereby show the scheme is an *extractable commitment scheme*, the simulator generates a key pair $(\mathsf{pk}, \mathsf{sk})$ for the crypto-system and sets the CRS to pk. Note the secret key sk now acts as a trapdoor that can be used to extract committed value m. ■

Therefore from Theorem 4.10 and Claim 4.10.1 we can state the following:

Theorem 4.11 (UC-OT from k-XOR and LPN)**.** *TThere are values of noise rate $\eta = \Theta(1)$, $k = \tilde{O}(\log r)$ and $m = r^{o(1)}$, such that there exists a UC-OT protocol if the following condition is satisfied:*

- *Constant-noise LPN is 2^{m^c}-hard (for any constant $c > 0$) and planted search k-XOR is $r^{k/2-o(1)}$-hard at every density $\frac{1}{\mathsf{polylog}(r)}$ where m is the dimension of secret used in LPN.*

CHAPTER 5

CONCLUSIONS AND FUTURE DIRECTIONS

5.1 CONCLUSION

- In this work, we saw a construction of Public-Key Encryption from somewhat mild hardness of planted search k-SUM at sufficiently low densities assuming weaker hardness of the Learning with Errors (LWE) problem than was known before.

- We then noticed that with small modifications to the LPN-based PKE scheme with k-XOR, we have a skeleton for a (Secret Key) Somewhat Homomorphic Encryption. However, some inherent limitations in this simple scheme were evident, i.e., the number of evaluations that can be supported and the number of ciphertexts that can be queried by an adversary in the underlying security game. We, therefore, introduced a new assumption that rooted its inspiration in the k-SUM problem and discussed a justification for it. This helped us allow the adversary to query (polynomially) unbounded number of ciphertext queries.

- We then used the k-XOR+LPN based PKE scheme to construct an Extractable Commitment scheme, which implies that we have a universally composable oblivious transfer protocol from “minicrypt" LPN and mild hardness of k-XOR.

- Finally, in the appendix, assuming hardness of k-XOR in the dense regime, we gave a construction of CPRF (Collision Resistant Hash Function) which can be used to replace traditional SIS based CRHF.

From all this, it is clear that assuming hardness of the k-SUM family of problems is contributing non-trivially in constructing advanced cryptographic primitives, and therefore, this warrants a deeper dive into the study of the hardness of k-SUM (and its variants). We also hope that this serves as a beginning to the exploration of applications of k-SUM in cryptography.

5.2 FUTURE DIRECTIONS

- It is worth noting that our SHE scheme still supports only a limited number of different ciphertext evaluations, and at present, we don't know of a way to increase the total number of evaluations possible beyond a certain limit. It would be interesting if we could somehow overcome this limitation.

- A limited amount of homomorphism can be useful in a few scenarios. For example, the setting of streaming algorithms. Therefore, it would be interesting to see if there are primitives/protocols in which our SHE scheme fits naturally. For example, we think this might help us build a sublinear MPC protocol as it might be possible to construct one using a limited homomorphic encryption scheme.

- We believe that there might be more primitives/protocols waiting that can utilize the hardness of the k-SUM problem and its variants in ways similar to PKE and SHE.

APPENDIX A

K-SUM IN THE DENSE REGIME

In this section, contrary to the main body of this work, we will look into one of the cryptographic applications of k-SUM (k-XOR) in the dense regime. To be precise, assuming hardness of k-XOR in the dense regime, we give construction of a CPRF (Collision Resistant Hash Function).

A.1 DEFINITIONS

Definition A.1 (Atmost-vector-k-SUM problem). Let $n, m = m(n), k = k(n), q = q(n)$ be integers and let $\Delta = \left\lceil \frac{k \log n}{m \log q} \right\rceil$. An algorithm $\mathcal{A}$ is said to solve the $\mathsf{avkSUM}(n, m, k, q)$ problem with success probability ϵ if, on input an instance $\mathbf{A}$ of size n it outputs an $S = \mathcal{A}(\mathbf{A}) \subseteq [n]$ with $2 \leq |S| \leq k$ such that:

$$\Pr_{S \leftarrow \mathcal{A}(\mathbf{A})} \left[\sum_{i \in S} \mathbf{a}_i = 0 \right] \geq \epsilon$$

where $\mathbf{A} \xleftarrow{\$} \mathbb{Z}_q^{m \times n}$ where $\mathbf{a}_i$ is the i^{th}-column of $\mathbf{A}$.

Conjecture A.1 (Atmost-vector-k-SUM conjecture)**.** *For any $n, k = \omega(1), m = \frac{k \log n}{3 \log q}$, any algorithm that solves* $\mathsf{avkSUM}(n, m, k, q)$ *with $1 - o(1)$ success probability has expected running time of at least $n^{3 \cdot o\left(\frac{k}{\log k}\right)}$.*

The above conjecture is justified in Section A.2.1[1]

Definition A.2 (Collision Resistant Hash Function). A *Collision-Resistant Hash Function (CRHF)* defined over a family of input spaces $\{\mathcal{X}_r\}_{r \in \mathbb{N}}$ and family of output spaces

[1]We use the word justification rather than proof because the precise statement works for $q >> 2$, but we use this to justify our belief in the hardness of the problem at any q.

$\{\mathcal{Y}_r\}_{r\in\mathbb{N}}$ consists of PPT algorithms $\mathsf{CRHF} = (\mathsf{Setup}, \mathsf{Hash})$ with the following syntax:

- $\mathsf{Setup}(1^r) \to s$: on input the unary representation of the security parameter r, outputs a key s (which describes a function $h : \mathcal{X}_r \to \mathcal{Y}_r$).
- $\mathsf{Hash}(s, x) \to y$: on input a key s and an input string $x \in \mathcal{X}_r$, outputs $y(= h(x)) \in \mathcal{Y}_r$.

It should satisfy the following properties:

Shrinking. CRHF is *shrinking* if for every security parameter r[2],

$$|\mathcal{X}_r| > \frac{|\mathcal{Y}_r|}{2}$$

Collision Resistance. CRHF is *collision-resistant* if for any PPT adversary A[3]:

$$\Pr[\mathsf{Hash}(s, x) = \mathsf{Hash}(s, x') \wedge (x \neq x')] \leq 1 - o(1)$$

where $s \leftarrow \mathsf{Setup}(1^r)$, $(x, x') \leftarrow \mathsf{A}(1^r, s)$ and $x, x' \in \mathcal{X}_r$

A.2 CRHF USING ATMOST-VECTOR-KSUM

Construction

Let $n \in \mathbb{N}, m = m(n), k = k(n) \in \mathbb{Z}$. Let input space $\mathcal{X}_n = \{\mathbf{x} \in \{0,1\}^n \mid ||\mathbf{x}||_0 = \frac{k}{2}\}$ and output space $\mathcal{Y}_n = \{0,1\}^m$. The CRHF is defined as follows:

[2]Two in the denominator can be replaced by any constant. And this non-trivial shrinkage is used instead of $|\mathcal{X}_r| > |\mathcal{Y}_r|$ so that we can use hardness amplification properties of CRHF

[3]It suffices to show that collision-resistance holds with probability at most $1 - o(1)$ instead of $\mathrm{negl}(r)$ because of the existence of a generic hardness amplification technique [CRS+07]

$\mathsf{Setup}(1^r)$: Upon input the security parameter represented in unary, do the following:

1. Set parameters $n = r$, $k = \omega(1)$, $m = \frac{k \log n}{3}$.
2. $\mathbf{A} \xleftarrow{\$} \mathbb{Z}_2^{m \times n}$.
3. Output $\mathbf{A}$.

$\mathsf{Hash}(\mathbf{A}, \mathbf{x})$: Upon input the function description $\mathbf{A}$ and the input $\mathbf{x} \in \mathcal{X}_n$, output $\mathbf{Ax}$.

Fig. A.2: CRHF construction

Theorem A.2. *Assuming conjecture A.1, the construction mentioned in Fig.A.2 is a CRHF.*

Proof. We prove the shrinkage and collision resistance property as mentioned in Definition.A.2 below. ■

Lemma A.3 (Shrinkage)**.** *For the construction mentioned in Theorem A.2, for every security parameter r*

$$|\mathcal{X}_n| > \frac{|\mathcal{Y}_n|}{2}$$

Proof. We know that $|\mathcal{X}_n| = \binom{n}{k}$, $|\mathcal{Y}_n| = 2^m$. Given $m = \frac{k \log n}{3} \implies \frac{\binom{n}{\frac{k}{2}}}{2^m} > 8 \implies |\mathcal{X}_n| = 8\,|\mathcal{Y}_n|$. ■

Lemma A.4 (Collision-Resistance)**.** *For the construction mentioned in Fig.A.2, for every security parameter r and any PPT adversary $\mathcal{A}$*

$$\Pr[\mathsf{Hash}(\mathbf{s}, \mathbf{x}) = \mathsf{Hash}(\mathbf{s}, \mathbf{x}') \wedge (\mathbf{x} \neq \mathbf{x}') \wedge (\mathbf{x}, \mathbf{x}') \in (\mathcal{X}_n)^2] \leq 1 - o(1)$$

where $\mathbf{s} \leftarrow \mathsf{Setup}(1^r)$, $(\mathbf{x}, \mathbf{x}') \leftarrow \mathcal{A}(1^r, \mathbf{s})$.

Proof. Assume there exists a PPT adversary $\mathcal{A}$ such that

$$\Pr[\mathsf{Hash}(\mathbf{s}, \mathbf{x}) = \mathsf{Hash}(\mathbf{s}, \mathbf{x}') \wedge (\mathbf{x} \neq \mathbf{x}')] = \epsilon$$

where $\mathbf{s} \leftarrow \mathsf{Setup}(1^r)$, $(\mathbf{x}, \mathbf{x}') \leftarrow \mathcal{A}(1^r, \mathbf{s})$ and $\mathbf{x}, \mathbf{x}' \in \mathcal{X}_n^2$.

Note that $\mathbf{A}$ is an atmost-vector-k-SUM instance over $\mathbb{Z}_2^m$ and if an efficient adversary $\mathcal{A}$ finds $\|\mathbf{x} \oplus \mathbf{x}'\|_0 \in [1, k]$ with probability ϵ then $\mathcal{A}$ solves the $\mathsf{avkSUM}(n, m, k, 2)$ over $\mathbb{Z}_2^m$ with probability ϵ. Therefore ϵ must be strictly less than $1 - o(1)$ by conjecture A.1 ∎

Applications of CPRF. Our CRHF can be used to replace the more traditional CRHF based on SIS (Shortest Integer Solution). For example, in the construction of hash encryption primitive, which implies IBE (Identity Based Encryption) mentioned in [DGHM17], we can use our CPRF instead of the SIS-based CPRF.

A.2.1 Justification for k-SUM family of Conjectures via Lattices

Lemma A.5 ([BSDV20])**.** *Let k, m, q, n be positive integers and $0 < \epsilon < \epsilon'$ where,*

$$q \approx k^{2(1+\epsilon')cN/\epsilon} \textit{ and } n \approx q^{\frac{\epsilon}{2\log k}} \textit{ for some universal constant } c > 0.$$

If there is an algorithm[4] *for average-case $\mathsf{kSUM}(n, k, q)$ that runs in time $T_{\mathsf{kSUM}} = T_{\mathsf{kSUM}}(n, k, q)$, then there is an algorithm for the worst-case $N^{1+\epsilon'}$-approximate shortest independent vectors problem (SIVP) that runs in time $2^{O\left(\frac{\epsilon N}{\epsilon' + \log N}\right)} T_{\mathsf{kSUM}}$ where N is the dimension of the underlying lattice.*

Corollary A.5.1. *Let $n, q = q(n), k = k(n) \in \mathbb{Z}^+$ where $k = \omega(1)$, $q >> k^{\frac{k}{3}}$, ϵ, ϵ' be constants. Then, assuming that the best known algorithm that solves $N^{1+\epsilon'}$-approx SIVP problem with probability of the order $1 - o(1)$ takes time $2^{\Omega(N)}$ for a constant ϵ', any algorithm that solves $\mathsf{kSUM}(n, k, q)$ with success probability $1 - o(1)$ must take atleast*

[4] We say an algorithm for kSUM or SIVP, if it solves with probability $\geq 1 - o(1)$

$n^{3 \cdot o\left(\frac{k}{\log k}\right)}$ *time.*

Corollary A.5.2. *Let* $n, m = m(n), q = q(n), k = k(n) \in \mathbb{Z}^+$ *such that* $k = \omega(1), \frac{k \log n}{m \log q} = 3, q >> k^{\frac{k}{3}}, \epsilon, \epsilon'$ *be constants. Then, assuming that the best known algorithm that solves* $N^{1+\epsilon'}$*-approx* SIVP *problem with probability of the order* $1 - o(1)$ *takes time* $2^{\Omega(N)}$ *for a constant* ϵ'*, any algorithm that solves* vector-kSUM(n, m, k, q) *with success probability* $1 - o(1)$ *must take atleast* $n^{3 \cdot o(\frac{k}{\log k})}$ *time.*

Proof. This directly follows from Corollary A.5.1 and the reduction given in Lemma 3.1 and the fact that for the parameters mentioned above $\frac{k \log k}{\Delta \log q} < 1$ ■

Justification for conjecture A.1

Claim A.5.1. Probability that there exist solutions to the vector-kSUM(n, m, i, q) problem for $i \in [2, \frac{k}{3})$ where $k = \omega(1)$ and $\Delta = 3$ is negligible in n.

Proof. Given $\mathbf{A} = (\mathbf{a}_1, .., \mathbf{a}_n)$ where $\forall j \in [n]$, $\mathbf{a}_j \xleftarrow{\$} \mathbb{Z}_q^m$.

$$\mathbb{E}[X] = \frac{\binom{n}{k}}{q^m}$$

where X denotes a random variable that counts number of subsets $S \subseteq \{\mathbf{a}_j\}_{j \in [n]}$ such that $|S| = i$ and $\sum_{\mathbf{a}_{j'} \in S} \mathbf{a}_{j'} = \mathbf{0} \mod q$.

Given $\mathbf{A} \in \mathbb{Z}_q^{m \times n}$, $k = \omega(1)$ and $\Delta = 3$, recall that $\Delta = \frac{\log \binom{n}{k}}{\log q^m}$ then for any $i \in [2, \frac{k}{3})$,

$$\begin{aligned}
\Pr[\textsf{vector-kSUM}(n, m, i, q) \text{ has solutions }] &= \frac{\binom{n}{i}}{q^m} \\
&= \frac{\binom{n}{i}}{\frac{\binom{n}{k}}{8}} \text{ (From } \Delta = 3) \\
&= \frac{8 \cdot \binom{n}{i}}{\binom{n}{k}}
\end{aligned}$$

$$
\begin{aligned}
\log\left(\Pr[\textsf{vector-kSUM}(n,m,i,q) \text{ has solutions }]\right) &= \log\left(\frac{8\cdot\binom{n}{i}}{\binom{n}{k}}\right) \\
&= \log\left(8\cdot\binom{n}{i}\right) - \log\left(\binom{n}{k}\right) \\
&\approx \log\left(8\cdot n^{i}\right) - \log\left(n^{k}\right) \\
&\approx \log\left(n^{i}\right) - \log\left(n^{k}\right) \\
&\approx (i-k)\log n
\end{aligned}
$$

Note that for any $i \in [2, \frac{k}{3})$, $(k-i) \geq \frac{2k}{3} \implies (k-i) = \omega(1)$

Therefore,

$$
\log\left(\Pr[\textsf{vector-kSUM}(n,m,i,q) \text{ has solutions }]\right) \approx -\omega(1)\log n
$$

$\implies$

$$
\Pr[\textsf{vector-kSUM}(n,m,i,q) \text{ has solutions }] \approx \frac{1}{n^{\omega(1)}}
$$

Because $k = O(n^c)$ for some constant c (it is an input to our PPT algorithms). Therefore, even if $\text{negl}(n)$ is added $\textsf{poly}(n)$ times, it is still negligible. Therefore, we conclude that the possibility of finding solutions for $\textsf{vector-kSUM}(n,m,i,q)$ for any $i \in [2, \frac{k}{3})$ is negligible in n. ■

Conjecture Justification: According to the corollary A.5.2, For any $i \in [\frac{k}{3}, k]$ the minimum time taken by an adversary that solves $\textsf{vector-kSUM}(n,m,i,q)$ is $n^{\frac{3\cdot c'\cdot i}{\log i}}$ for some constant c',which is the lowest for $i = \frac{k}{3}$ but this is still greater than $n^{o\left(\frac{\log k}{k}\right)}$. Hence, by corollary A.5.2 and Claim A.5.1, any PPT adversary will not be able to solve $\textsf{avkSUM}(n,m,k,q)$.

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